Quantifying Narrative Diffusion Across Languages

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Abstract

How can one trace the diffusion of ideas across texts? This is a key measurement problem in sociological inquiry, from the study of how interest groups shape media discourse (Bail, 2012, 2015; Wetts, 2023), to the spread of policy across institutions (Hinkle, 2015a), to the diffusion of organizational structures and institutions themselves (Meyer and Rowan, 1977). To measure diffusion, we seek to identify texts that make the same set of specific claims about the same events, a phenomenon we call "narrative diffusion." We use instruction-trained LLMs to distill texts to their core ideas and then compare them at the level of the *claim* instead of words, phrases, or sentences. The result is an estimand much closer to "narrative diffusion" than is possible with exact text matching (e.g., Bail, 2012, 2015; Hinkle, 2015a; Wetts, 2023), which returns lexically similar documents, or topic modeling (e.g., Ceron et al., 2021a; Krawczyk et al., 2021), which returns topically similar documents. We apply our approach to an important case study: the spread of Russian claims about the development of a Ukrainian bioweapons program in U.S. mainstream and fringe news websites. While we focus on news in this application, our approach can be applied more broadly to the study of propaganda, misinformation, diffusion of policy and cultural objects, amongst other topics.

1 Introduction

The digital revolution has upended the 20th century's media ecosystem (e.g. Abernathy, 2018), creating massive downward cost pressures on conventional outlets (Saridou et al., 2017; Boumans et al., 2018), and thus opportunities for state-funded foreign language media sources. While these sources—including Russia's RT, Tass, and Sputnik—reach only a small English-language audience directly, they can sometimes "launder" Russian state messages via "pickup" coverage in conventional Western outlets (Miskimmon and O'loughlin, 2017; Oates and Ramsay, 2024). For example, Oates and Ramsay (2024, pg 154) provide examples of U.S. media reprinting false conspiracies about neo-Nazi influence in the Ukraine government in 2022, a Russian state media narrative originating from Vladimir Putin himself. The authors found dozens of stories from U.S. right wing websites that repeated the claim—some presented it as truth, while others reported the claim in quotes from Putin, but did little to debunk it. They found even more mentions of the claim in mainstream news outlets, although it was often debunked.[1]

This is just one example of a phenomenon we call "narrative diffusion," whereby content spreads across the media ecosystem, often among legitimate actors, and sometimes without attribution or context. In the case of state propaganda, narrative diffusion is not just a function of the actions of malign actors, but also of the pressures in contemporary journalism to cover global events with limited and shrinking resources and the ease of narrative diffusion in the digital information environment (Boumans et al., 2018; Cagé et al., 2020; Nicholls, 2019; Saridou et al., 2017).

Narrative diffusion is closely linked to an extensive literature across sociology, political science, and communication which attempts to measure the diffusion of ideas in text media. These diverse works apply measures of information influence and diffusion to understand a range of empirical phe-

[1]These narratives are often amplified on social media accounts. In 2017, among 600 Russian social media accounts tracked by the German Marshall Fund (see `https://securingdemocracy.gmfus.org/hamilton-dashboard/`), the most mentioned news organizations included Russian propaganda outlets RT and Sputnik, along with U.S. media outlets like Fox News, the Gateway Pundit and Breitbart. See Dorell, Oren. "Breitbart, other 'alt-right' websites are the darlings of Russian propaganda effort." *ABC News* August 24, 2017. `https://www.usatoday.com/story/news/world/2017/08/24/breitbart-other-alt-right-websites-darlings-russian-propaganda-effort/598258001/` Accessed. August 6, 2024.

nomenon, including the evolution and flow of policy ideas (Callaghan et al., 2020; Hinkle, 2015b; Kroeger et al., 2022; Wilkerson et al., 2015), how media agendas are set (Welbers et al., 2018; Tsur et al., 2015), how information flows are shaped by censorship (Lu et al., 2024), how interest groups shape media discourse Bail (2012, 2015); Wetts (2023), how the media cover members of congress (Grimmer, 2013), and patterns of "churnalism," in which journalists reuse each other's content (Boumans et al., 2018; Cagé et al., 2020; Nicholls, 2019; Saridou et al., 2017). This rich literature on the flow of information and ideas over time and across organizations can be applied to the study of propaganda—and narrative laundering in particular—which motivates both our methodological and empirical work presented below.

We proceed as follows. First we define narrative diffusion and discuss how this concept has previously been measured. Second, we introduce our data and discuss our measurement strategy. Third, we overview the steps we took to validate our method and compare it to existing approaches. Finally, we present an empirical application using our approach: tracking diffusion of narratives related to Russian claims that Ukraine was operating a US-funded "biolab." While we focus on narrative diffusion specifically for our substantive case, our framework could be used to studying instances of information diffusion and reuse more broadly.

2 What is Narrative Diffusion?

What is a narrative and why do we use the term "narrative diffusion"? Past work defines narratives as representations of events or sequences thereof, temporally and/or causally linked (Ryan, 2007). Narratives are used to tell a *story* (Rudrum, 2005) about an underlying sequence of events, creating a intelligible plot out of an otherwise disparate set of actors and actions (Somers, 1994; Polletta et al., 2011). While sociological work on narratives has often focused on the stories individual actors tell about their life histories, identity, and decision making (Kiviat, 2019; Frost, 2019; Somers, 1994), the concept of narrative is also important in the sociology of news, and is closely related to how newsmakers frame events to their audiences (Bail, 2012; Fiss and Hirsch, 2005; Gamson and

Modigliani, 1989).

Recent work in international relations has extended the concept of narrative to *strategic narratives*, or the intentional deployment of narratives to further organizational and state objectives (Miskimmon et al., 2014). Research has especially focused on how revisionist powers such as Russia and China are using control over their media systems to generate "soft power" influence campaigns abroad (Szostek, 2017; Khaldarova and Pantti, 2020; Roselle et al., 2014; Guo et al., 2019; Fan et al., 2024) and attempt to interfere in external media environments (Golovchenko et al., 2020; Eady et al., 2023). Russia for example has invested significant resources into its global media operations, generating content in over many languages in state media outlets such as Russia Today (RT) (Elswah and Howard, 2020; Redington, 2021).

This literature on strategic competition over narratives in the media environment has created a broad interest in *quantitatively* estimating the extent to which different narrative diffuse across contexts and thus the extent to which countries such as Russia have been successful in their media influence campaigns. Our work fits within these efforts. Drawing on this conception of narratives as stories and representations of underlying events, actors, and objects, we define a narrative as a claim, argument, or framework focused on a *specific* phenomenon/event/subject, or a set thereof. We consider two texts to share the same narrative if they make the same claims about the same underlying event. These claims can be descriptive (a statement about what is), normative (a statement about what should be), and classificatory (a statement about what something is a case of).

In addition to being faithful to the idea of narrative, using this specific definition of narrative allows us to estimate *narrative commonality* across texts in a corpus. *Narrative diffusion* is a process that creates *narrative commonality* as an empirical relic: documents discuss the same narrative—they make the same claims about the same phenomenon, subject, or set of events. Patterns of diffusion can then be studied using longitudinal and/or network analyses on data describing narrative commonality.

Previous work on identifying narratives at scale, however, suffers from a range of shortcomings which make it insufficient for our goal to measure the diffusion of narratives across texts. One

set of work has focused on shared, common frames across a large number of documents. For example, Ash et al. (2024) uses semantic role labelling, embedding, and clustering methods to identify relationships between named entities in document.[2] Similarly, Wang and Rambow (2024) extract and cluster [*source, target, factuality*] triplets at the sentence level to predict "influence campaigns," which they show performs better than document-level prediction. However, these approaches do not capture the specificity of narratives and events from our definition. The fact that there are cultural, material, physical constraints on story telling (Polletta et al., 2011), leading to common and shared narratives or stories about different events (Polletta, 2015), should not distract from the fact that narratives are representations of unique underlying sequences.

A second approach that has been applied specifically to narrative extraction and the study of information flows more broadly is topic modeling (Ng et al., 2021; Ghasiya and Okamura, 2021; Ceron et al., 2021b; Krawczyk et al., 2021; Hanley et al., 2023; Madrid-Morales, 2021). Topic models can group documents together that are on the same topic even if the words used differ. However, as an estimator, topic models are disconnected from the actual patterns of diffusion in media, instead assuming that text is generated from a topic in a stochastic process. Topic models cluster text on the same topic, without distinguishing between more subtle claims and specific events—topic models do not capture argumentation, evidence, or even valence. One can share themes without talking about the same narrative.

Finally, exact text matching has been used to reveal dependent patterns of shared information (Boumans et al., 2018; Cagé et al., 2020; Nicholls, 2019; Saridou et al., 2017), but is also not a suitable estimator for our purposes. Exact text matching and text reuse methods require that the writer copies some amount of text verbatim from an original document. The advantage of this approach is that if one document shared substantial textual overlap with another it is unlikely to be driven by chance. This approach, however, will miss instances of writers paraphrasing one another and other instances of writers extracting key information and putting it in their own voice. The assumption that overlapping exact text phrases can capture narrative diffusion is especially

[2]For an extensive review of narrative extraction approaches which similarly focus on shared themes across documents, see Keith Norambuena et al. (2023).

problematic in the study of information copying in media sources, in which ethical standards and copyright law prohibit direct text reuse without agreement. What's more, exact text copying limits informational reuse methods to monolingual applications, even as cross-linguistic reuse has become more common with the availability of cheap translation technologies. One can copy an idea without copying the exact text.

Any attempt to *measure* narrative commonality and diffusion must carefully consider the *estimand* and the *estimator* thereof (Lundberg et al., 2021). We use an approach that more closely maps onto our underlying target and compare the performance of this approach against human generated labels and alternative approaches. Specifically, we use large language models (LLMs) to compare documents to each other and assess whether each document pair is in fact making the same *specific* claims about the same *specific* events. We use LLMs simply because—unlike all past natural language technologies we have encountered—instruction-tuned LLMs have proven capable of completing this task at a level that is impressive when quantified (see validation section below).

While this approach solves many problems, it requires the use of additional methods to scale to large corpora. If we were to compare all of the articles with each other, the number of pairwise comparisons and thus computing required increases sharply (at a rate of $\frac{n^2-n}{2}$). Thus, to identify articles sharing the same narrative, we proceed in three steps designed to navigate these tradeoffs and scale our method. First, we use LLMs to summarize (and translate when necessary) articles based on "chain of thought" to distill texts to the core claims and subjects (Liu et al., 2022; Wei et al., 2023; Kojima et al., 2023; Zhou et al., 2023; Wu et al., 2023). Second, we generate "candidate pairs"—a set of document pairs that have a higher likelihood of actually making a similar claim. We identify these candidates using a pre-trained Bert-based sentence transformer model to generate a semantic similarity score between document pairs (Reimers and Gurevych, 2019). And finally, we develop a prompt for an instruct-tuned LLM to identify pairs of documents that contain the same claims on the same topics, applying this prompt to each of the candidate pairs.

Our approach has parallels with a number of existing literatures in NLP and text as data. It is closely connected with the paraphrasing literature in computer science, where the motivating

task was historically plagiarism detection. Of course, plagiarism can be more sophisticated than exact string-matching allows, and these methods attempt to measure similarity of meaning across words and sometimes sentences by employing semantic similarity or semantic relatedness (Vrbanec and Meštrović, 2020). These methods are now used in applications beyond plagiarism detection, including natural language understanding and information retrieval. However semantic relatedness is different from similarity of claims as perceived by actual language speakers. As we show below, pairs of texts may score high on semantic relatedness while still not talking about the exact same claims and events. Our quantity of interest is also closely related to text matching (Roberts et al., 2020; Mozer et al., 2020), although with the caveat that we want to identify all pairs of articles which "match" rather than the single, closest pair.

We emphasize that *documents that contain the same narratives* provide evidence of narrative diffusion—they are an empirical byproduct out of which we can generate useful data. However, we cannot know what exactly causes similar sets of claims to appear in different outlets. Nevertheless, that measure of narrative commonality can be combined with temporal precedence to draw narrative connections between texts.

We further emphasize that our method will often draw connections between texts providing unbiased coverage of important, newsworthy real-world events. Thus, to study the diffusion of misinformation, propaganda, and other content likely to depart from the mainstream of descriptive claims related to ebb and flow of everyday news events, it may be fruitful to pair this method with data regarding sources and/or topic models, and of course careful reading and examination of the resulting data (Grimmer and Stewart, 2013), as we seek to do in the application below.

3 Data

In this paper we draw on a corpus of news website articles we created to estimate narrative diffusion in the U.S. media environment during the full-scale Russian invasion of Ukraine in February 2022. Our data comprises a multi-lingual corpus of news articles collected from forty-five public

news websites (see table 1). These include four Russian state media sources, eighteen U.S. popular mainstream news websites, eleven low quality U.S. news websites, and twelve Ukrainian news websites. We chose this balance of news websites because we wanted to investigate which types of news websites in the United States might be promoting Russian narratives, a hypothesis we return to in the application section below. The four Russian sources are the main state-owned media sources in Russia for domestic and international audiences (Tass, Pravda, Russia Today, and Sputnik Henriksen et al., 2024). The eighteen popular mainstream news websites are among the top twenty-five news websites by consumption in the United States.[3] The low quality news websites are among the 569 websites previously identified by (Allcott et al., 2019) as "fake news" websites. Finally, the Ukranian media sources are high quality, mainstream news sources previously identified by Aslett et al. (nd).[4]

The dataset includes 692,560 articles published between January 1, 2022 and April 30, 2022, a two-month window on either side around the full scale invasion on February 22, 2022. We created this dataset through a rigorous process of collecting article URLs, parsing the raw HTML from article web pages into structured text fields, and cleaning the dataset to remove duplicates and articles with less than 200 characters.[5] We conducted a rigorous validation exercise to test for missigness in our data. Out of 49 sources (breaking out different languages of the same source), 34 sources had a missingness rate of ten percent or less (69%) and only 7 sources had a missingness rate greater than 15%.[6]

Source	**Domain**	**Type**
100 Percent Fed Up	English	US low quality
ABC News	English	US popular mainstream
Bipartisan Report	English	US low quality
Business Insider	English	US popular mainstream
Bykvu	English, Russian, Ukrainian	Ukrainian
CBS News	English	US popular mainstream

[3]These were identified by (Aslett et al., nd) using Microsoft Research's Project Ratio. These news websites represent the top sites by consumption from 2016 to 2019. `https://www.microsoft.com/en-us/research/project/project-ratio/`

[4](Aslett et al., nd) used the independent journalism cite Texty to identify these sources. `allcott2019trends`

[5]See the Supplemental Materials for further details on how we created this dataset.

[6]See Supplemental Index, Section B for more details.

Table 1 continued from previous page

Source	Languages	Type
Censor.net	English, Russian, Ukrainian	Ukrainian
Clash Daily	English	US low quality
CNBC	English	US popular mainstream
CNN	English	US popular mainstream
Daily Caller	English	US low quality
Fakty.ua	Russian, Ukrainian	Ukrainian
Fox News	English	US popular mainstream
Gordon	English, Russian, Ukrainian	Ukrainian
HuffPost	English	US popular mainstream
IJR	English	US low quality
Infowars	English	US low quality
Interfax	English, Russian, Ukrainian	Ukrainian
LB.ua	English, Ukrainian	Ukrainian
MSNBC	English	US popular mainstream
Natural News	English	US low quality
NBC News	English	US popular mainstream
New York Post	English	US popular mainstream
New York Times	English	US popular mainstream
NPR	English	US popular mainstream
NV.ua	Russian, Ukrainian	Ukrainian
Palmer Report	English	US low quality
PBS.org/newshour	English	US popular mainstream
Politico	English	US popular mainstream
Pravda.ru	English, Russian, Ukrainian	Russian state media
RBC.ua	Russian, Ukrainian	Ukrainian
RT (Russia Today)	English, Russian	Russian state media
Slate	English	US popular mainstream
Sputnik	English	Russian state media
Stillness in the Storm	English	US low quality
Tass	English, Russian	Russian state media
The Gateway Pundit	English	US low quality
The Hill	English	US popular mainstream
The Political Insider	English	US low quality
TSN.ua	English, Russian, Ukrainian	Ukrainian
Ukrinform	English, Russian, Ukrainian	Ukrainian
Unian	Russian, Ukrainian	Ukrainian
USA Today	English	US popular mainstream
Washington Post	English	US popular mainstream
ZN.ua	Russian, Ukrainian	Ukrainian

Table 1: Sources in dataset, including main language variations included and media outlet type.

For a number of reasons we restrict our validation and application analysis to a subset of these articles, focusing on a single case: articles related to rumors that Ukraine was operating US-funded

bioweapons. We do so in spite of the limitations this creates for generalizability. The main reason for doing so over randomly sampling relates to the pairwise nature of our approach—randomly sampling would introduce sparsity into the networks of narrative diffusion, severely limiting the occurrence of positive matching document pairs. Focusing in on a census of documents within a relatively contained content network allows us to evaluate the approach in a setting that allows for a relatively higher rate of positive matches.[7] What's more, a larger corpus would create additional computational challenges, and given the difficult nature of this task, we wanted to start with a relatively small corpus first. Finally, as we discuss below, this case is of substantive interest—allegations of Ukrainian bioweapon development may be used by Russia to justify the use of weapons of mass destruction (WMD).

On March 6, 2022, less than two weeks after Russia's full-scale invasion of Ukraine, the Russian Ministry of Defense published what it called evidence of a "military biological program financed by the US Department of Defense in Ukraine." The ministry claimed that it had learned from employees of Ukrainian biological laboratories that hazardous pathogens such as plague and anthrax had been destroyed immediately before the invasion to conceal work on biological weapons. Over the following months, Russian officials unveiled additional allegations, claiming, e.g., that the alleged U.S. "biolabs" in Ukraine developed pathogens targeting specific ethnic groups or that Ukraine was preparing to attack Russia with infected birds and bats (Editorial Board, 2023). On March 11, March 18, and May 13, Russia called for U.N. Security Council meetings to discuss its accusations. On March 16, Russian President Vladimir Putin repeated these claims. On March 24, Russia launched another series of allegations, accusing President Biden's son Hunter of securing funding for the "bioweapons program." TASS, RT, Sputnik, RIA Novosti, and other Russian state outlets extensively covered the "biolabs" claims throughout the spring of 2022.

The Russian allegations were, as reputable news agencies and independent fact-checkers have demonstrated (Cercone, 2022; Kessler, 2022b,a), false. [8] The documents and other "evidence" that

[7] However, this rate is still very sparse—on the order of 1 in 300 positives).

[8] While the U.S. has been indeed supporting biological research in Ukraine and other post-Soviet countries, these efforts under the umbrella of the Biological Threat Reduction Program (later the Cooperative Biological Engagement Program) have been focused on increasing biological safety and security and preventing the weaponization or mishan-

Russia provided were either fabricated or grossly misinterpreted by Russian officials and media. The 2022 disinformation campaign was rooted in earlier Russian and Soviet disinformation efforts (Roffey and Tunemalm, 2017; Leitenberg, 2020).[9] Since 2014, as Russia annexed Crimea and started a military conflict with Ukraine, the Ukrainian government has been increasingly featured in "bioweapons" claims. According to a 2021 study, for example, pro-Russian Ukrainian media circulated claims that the U.S. had set up biological laboratories in the country to experiment on Ukrainians. The allegations that Russia revealed in March 2022 could have been pre-planned by the Kremlin to justify the new phase of the conflict, building on earlier propaganda efforts. At the same time, claims about U.S.-funded Ukrainian "biolabs" also started circulating in U.S. social media the day after Russia's full-scale invasion on February 24 (Cercone, 2022). These allegations may have revealed that there was a demand in the West for such conspiracies, prompting the Kremlin to launch a more extensive and sophisticated disinformation campaign around Ukrainian "bioweapons."

This series of events and incidents is a great case study to test the ability of our method to identify the diffusion of narratives from Russian state media to US media sources. We focus on this case because it represents an ideal setting for the types of mechanisms that might generate narrative diffusion. International media organizations wouldn't have had the resources or access to "on the ground" developments related to the war and bioweapons controversy. Russia has invested significant resources in its propaganda apparatus, including its state media outlets (Paul and Matthews, 2016), and has incentives to influence media coverage by other sources. Indeed anecdotal evidence suggests that these state outlets have previously been effective in generating favorable narratives outside of Russia (Ramsay and Robertshaw, 2018; Oates et al., 2020; Watanabe, 2017). The invasion also prompted concerns about U.S. media's laundering of Russian narratives (Messieh, 2023), which we investigate with the methods developed in this paper.

To identify articles in our multilingual corpus related to the bioweapons case study, we filter our

dling of biological materials.

[9]The specific allegations about the U.S. "biological weapons" program in the former Soviet space have been raised by Russian officials at least since the early 2010s. Initially, they were focused more on Georgia, which the Kremlin's disinformation campaigns vilified in the wake of the Russo-Georgian War of 2008.

dataset to 3,491 articles which contain keywords related to "bioweapons" and the word "ukraine."[10] We also restricted the dataset to articles published between January 1st and May 1st of 2022.

4 Method

In this paper we develop a method to measure narrative diffusion that more directly maps onto the underlying phenomenon than previous approaches (Lundberg et al., 2021), whether based on exact text matching (Boumans et al., 2018; Cagé et al., 2020; Nicholls, 2019; Saridou et al., 2017) or topic modelling (Ng et al., 2021; Ghasiya and Okamura, 2021; Ceron et al., 2021b; Krawczyk et al., 2021). As noted in the introduction, we argue that narrative diffusion occurs at the level of document pairs and thus must be measured at that level, rather than identifying common themes measured in the aggregate across documents. This pairwise approach, however, creates problems at it solves others, as following it requires performing pairwise operations across documents, which quickly grows in computational complexity on the order of $\mathcal{O}(n^2)$.[11] In just our dataset of 3,491 articles for our case study, there are over 6 million unique pairs of articles.

Our estimator includes three steps to meet these twin challenges of pairwise comparison and measuring narrative diffusion. First, drawing on insights from a literature highlighting "chain of thought" prompting (Wei et al., 2023), we summarize and extract the key features of each document. Second, we reduce the number of pairwise comparisons from over 6 million to approximately 60,000 in a "candidate selection" process utilizing tools from semantic similarity. Finally, we use LLM annotation to compare the claims and subjects in the remaining 60,000 pairs to identify pairs of documents that make the same claim(s) about the same subject(s).

Although pairwise modeling approaches mean samples are no longer independent, the computer science literature on data augmentation suggests pairwise modeling can increase classification ac-

[10] The biolabs keywords we used were "biolab", "biological", "bioweapon", and "pathogen", with their equivalents in Russian and Ukrainian.

[11] Throughout we use "Big O" notation to describe computational complexity. This notation is commonly used in the computer science literature to describe the asymptotic properties of an algorithm—particularly run time or space—as a function of input.

curacy by "virtually increasing sample size". For example, in computer vision, paired images have been combined to increase the total number of training samples, which has been found to reduce cross-validated error rates (Inoue, 2018). Other examples can be found in Tynes et al. (2021).

4.1 Chain of Thought Summarization

We first extract the core claims and subjects in a document in order to impose a degree of uniformity in language, formatting, length, style, and overall number of claims, which greatly reduces the effective dimensionality[12] of the various downstream modeling tasks, and increases the performance of the overall method. We do so using an LLM prompting approach called "chain of thought" (Wei et al., 2023; Wu et al., 2023). Chain of thought explicitly prompts an LLM to generate intermediate reasoning necessary to produce a final judgement or output. There is a fast-growing emergent literature showing that generative LLMs require and perform better with this iterative step-by-step structure to minimize errors (Liu et al., 2022; Wei et al., 2023; Kojima et al., 2023; Zhou et al., 2023).

We use an instruction-tuned LLM to summarize and distill the core claims and subjects of our articles in a two-step process. First, we prompt the LLM—in our case GPT4o—to summarize each text in 5-7 English sentences. We thereby distill the points of each text into documents of standardized length and also ensure that our corpus is all English. The language standardization is necessary to avoid between language differences in performance in the second and third steps of our approach.

Second, in two separate prompts, we prompt GPT4o to extract the "descriptive, normative, causal, and classificatory claims" (the "claims") and "people, places, things, and events" (the "subjects") included in each summary. Our prompts instruct GPT4o to list these claims and subjects. These definitions of subject and claim mirror how we define narrative diffusion in our discussion above. Rather than passing the full article text or article summary, it is these lists that we prompt

[12]To see this, consider the total number of bag-of-words features before and after this transformation. Featurizing text in a single language with fewer unique words will translate to fewer features.

the LLM to compare in the final step or our method.

4.2 Generating Candidate Pairs

Comparing claims requires performing pairwise operations across all relevant documents, which grows on the order $\mathcal{O}(n^2)$. To reduce the number of comparisons we need to make between each pair of article's claims and summaries we use a computationally inexpensive method to generate "candidate pairs", or pairs of articles that are potentially making similar claims about the same subjects. Our goal in this step is to filter out the most number of irrelevant pairs as possible while still recalling all actual cases of narrative commonality.

We identify pairs that are more likely to be engaging in narrative reuse with sentence transformers or SBERT (Reimers and Gurevych, 2019). SBERT (Reimers and Gurevych, 2019) is an extension of the BERT language model, explicitly designed with pairwise semantic similarity tasks in mind. SBERT supports two basic approaches for comparing sentences (or paragraphs), both of which we use here. The first is called a bi-encoder approach, in which the user generates separate embeddings for each text and identifies similar texts using a distance metric like cosine similarity. Second, SBERT supports a cross-encoder approach, in which the user jointly encodes each input text using the BERT transformer network and produces a single similarity score for each pair. The cross-encoder approach is orders of magnitude more computationally costly,[13] but generally produces superior results (Reimers and Gurevych, 2019). As recommended in the literature, we combine these approaches in our candidate generation step. We first generate embeddings for each of the article summaries (not the list of claims and subjects) in our dataset and compare the distance between all embedding vectors in our data set.[14] With a cutoff tuned on our recall set (discussed below), we reduce the number of potential positive matches from 6.09 million to 392,320 with the

[13]The cross-encoder is still orders of magnitude less costly than prompting an instruction-tuned LLM.

[14]We calculate the distance using cosine similarity, which SBERT embeddings were designed to support. SBERT supports a range of models. We use the mpnet base model ("all-mpnet-base-v2"), as this is SBERT's currently reported high performing general purpose model `https://sbert.net/docs/sentence_transformer/pretrained_models.html`. Accessed July 31, 2024. See Supplemental Index Section E for a comparison between MPNet and another top performing semantic similarity model, STS Roberta Large. Our approach in this step is similar to the approach by (Hanley et al., 2023), who also uses MPNet embeddings to cluster texts.

bi-encoder step, allowing us to reduce the number of pairwise comparisons by over 93%.

In the second and final step of our candidate generation process we pass the 392,320 pairs returned by the bi-encoder through an SBERT model cross-encoder. We then tune this between-text semantic similarity score to create a cutoff for inclusion in the candidate pair set. In the validation section below we discuss the performance of this cutoff. With this step we reduce the number of candidate pairs from 392,320 to 64,677 pairs. The candidate process ultimately discards 6,027,118 out of 6,091,795 pairs, or 98.9% of all possible pairs in the case study.

4.3 LLM Annotation

The final step of our narrative diffusion method is to use an instruct-tuned LLM to directly compare and annotate the lists of claims and subjects for our candidate pairs. In the section that follows we evaluate the performance of our estimator against a gold standard human evaluation procedure. In order to map onto this procedure as closely as possible, we developed prompts that reflected the code book that we gave to our human coders. In particular, for each candidate pair's list of subjects and list of claims we (separately) prompted the following two prompts:

- **Same Subject**: You will be provided with the lists of the people, places, objects, and events discussed in two paragraphs. Based on these lists, do the two paragraphs discuss the vast majority of the same people, places, objects, and events? List for Paragraph 1: [insert list] List for Paragraph 2: [insert list] Your Label (Respond only 'YES' or 'NO')

- **Same Claim**: You will be provided with the lists of descriptive, normative, conceptual, and causal claims discussed in two paragraphs. Based on these lists, do the two paragraphs discuss the vast majority of the same claims? List for Paragraph 1: [insert list] List for Paragraph 2: [insert list] Your Label (Respond only 'YES' or 'NO')

We pass the appropriate lists (subject or claim) from each of the 64,677 candidate pairs once through each prompt. We used GPT4o for our model. We label pairs of articles which were annotated

"YES" for same subject and same claim as predicted positive cases, i.e. pairs wherein our estimator detected common narratives.

While our final prompts largely reflect our human codebook, we tried a number of different strategies before finalizing these prompts. First, consistent with recent work showing zero-shot learning can outperform few-shot learning (i.e., an LLM performs better without providing any examples than providing just a few Reynolds and McDonell, 2021; Kojima et al., 2023), we find that removing examples from our set of instructions improved performance.[15] We also found that providing the lists of subjects and claims using Markdown (a lightweight markup language for formatting text) avoided confusion between the two. We also set the model "temperature"—a parameter governing randomness and thus diversity in next word prediction—to zero, to help ensure the model would choose the highest probability label.[16] We also edited the instruction text for brevity, since the context window for instruct-tuned LLMs is finite. Finally, we used a stricter standard for matching than we used for human coders, prompting the LLM to evaluate whether the two lists discuss the *vast majority* of the same subjects or claims.[17]

Pseudocode describing our SBERT-LLM approach follows:

4.4 Alternative Measures

In the validation section below we compare the performance of our narrative estimator with approaches based on topic modelling and exact text reuse. We test a text reuse approach by measuring the between article 5-word gram cosine similarity for all articles in our bioweapons case study. 5 word-gram cosine similarity varies from 0 to 1, where 0 indicates the 5-gram vectors of a pair of articles have no overlapping features and 1 indicates the two vectors have the same features up to a scalar multiple. This is a common approach in the text reuse literature. For example (Callaghan et al., 2020; Kroeger et al., 2022) uses 5-word gram Jaccard similarity to classify the origin of state

[15]The authors of Reynolds and McDonell (2021) propose that "the function of few-shot examples in these cases is better described as locating an already learned task rather than meta-learning

[16]The temperature parameter has been compared to the model's "creativity" (Peeperkorn et al., 2024)

[17]We asked our human coders whether the two lists shared "more than the majority." We used this stricter standard because we found on average our human coders were more strict with their evaluations.

Algorithm 1 Estimating Narrative Commonality

```
for ∪{ego_text, alter_text} = 1, 2, ..., (N + M) do
    Summarize text as S_text
    Extract lists of claims and subjects from S_text as C_lists
end for
for ego_text = 1, 2, ..., N do
    for alter_text = 1, 2, ..., M do
        SBERT bi-encode (ego_S_text, alter_S_text) as sbert_bi_score
        if sbert_bi_score > threshold_bi then
            SBERT cross-encode (ego_S_text, alter_S_text) as sbert_cross_score
            if sbert_cross_score > threshold_cross then
                Compare C_ego_lists, C_alter_lists via instruct-LLM prompt
            end if
        end if
    end for
end for
```

legislature bills (whether copied from the model bills of interest groups, other state legislatures' bills, or developed internally). (Nicholls, 2019) uses 5 and 7-word Jaccard similarity to measure newspaper articles' reuse of copy from news wires, press releases, and other outlets. Following (Waight et al., 2024), we use cosine similarity rather than Jaccard similarity because it is less sensitive to document length.

We also evaluate the performance of a topic model approach by placing documents into coarsened topic clusters. We fit a structural topic model (Roberts et al., 2014, 2019) to our bioweapons case study documents. We use a topic model with 30 topics, selected based on substantive meaningfulness of the model topics and quantitative diagnostics of semantic coherence and topic exclusivity.[18] In order to group articles together which have similar topic distributions, we coarsened each document's topic prevalence vectors into 1-0 bins. Each document was thereby represented as a vector of binary variables, where documents had a given topic indicator when their topic prevalence estimate for a given topic was above .2 (meaning 20% of document words were estimated to be drawn from that topic's distribution over words). We then grouped documents into clusters of documents that have the same unique combination of binned topics.[19] This approach builds off of

[18] See SI Section D for comparisons between models on these estimates.

[19] We chose a threshold of .2 because above that threshold a rapidly increasing number of documents had no topics at the threshold and below that threshold a rapidly increasing number documents were not placed into any topic cluster.

text matching approaches using coarsened topic representations (Roberts et al., 2020). For both the text reuse estimator and the topic clustering estimator we translated any non-English documents into English using Google Translate and ran the estimators on the full document translated text.

5 Validation

Validating any classifier where positive cases are rare is difficult, as is the case here. The problem lies in finding a well-powered but low-bias sample of ground truth positives. For example, if there is 1 positive case for every 1000 documents, coders would naively need to code a random sample of 100,000 documents to generate a sample of 100 ground truth positive examples. By virtue of the way newspapers work only a small fraction of texts will even be on the same topic, much less contain identical claims. And, the pairwise nature of this task compounds this positive-sparsity further due to the steep rate at which the number of potential matches increases as the number of documents grows ($\frac{n^2-n}{2}$). Out of the 3,491 articles in our case study, we estimated with our LLM-based approach that 18,318 pairs of articles were making the same claim about the same event, or .3% of approximately six million unordered pairs. We estimate that a human coder would need to read 300 randomly sampled pairs to identify at least one of these cases. This issue of sparsity also makes this problem particularly ill-suited for machine learning approaches that require representative training data.

This situation creates challenges for estimating classifier recall, or how well the classifier detects true instances of the phenomenon in question—in our case common narratives. Recall is formally the "true positives" classified by the model over all ground-truth-positive cases (true positives (TP) plus false negatives (FN)), $\frac{TP}{TP+FN}$. Sparse data makes it challenging to identify these positive cases.

To generate a sample of ground-truth-positive cases, we attempt to identify cases near the decision boundary rather than randomly sampling pairs. This greatly reduces the number of cases that need gold standard human-coded labels (see for example, Settles, 2009). To do so, we employ a series of ranking algorithms to generate a set of documents with higher-than-random semantic

similarity. In particular, we start with a random sample of "seed articles" from Russian state media. We rank articles from other sources in our dataset by their estimated degree of semantic similarity (specific details below) with the focal Russian state media article and then have humans code these pairs in ranked order in terms of narrative commonality defined above (similar to the approach used in Lu et al. (2024) to generate human coded pairs).

We use two different ranking algorithms to create these ranked sets. First, we rank potential matches with each of the seed articles by the cosine distance between each article pair's vectors of simple bag-of-words counts over the full article, weighted by TF-IDF (tf-idf BoW cosine similarity).[20] Second, we rank potential matches by the cosine distance between each article pair's embedded vector representation based on the Universal Sentence Encoder (USE). In order to be within the maximum context length for USE, we embedded each article in the USE embedding space after summarizing each article in English using GPT4o. Pairs of articles which rank highly in USE cosine similarity will be closer in the semantic space as defined by the USE model.[21] Given that both ranking algorithms rely on some architecture from the estimators we are trying to evaluate (TF-IDF BoW on exact text featurization, USE on GPT4o summaries), we randomize the ranking algorithm we use for each focal article's ranked set of potential matches in order to avoid selecting a recall set that favors one approach over the other.[22]

Research assistants coded USE or BOW ranked lists for 250 focal articles randomly selected from 959 Russian state media articles in the bioweapons case study. We limited potential matches to those that were published within a five-day window from the date of publication of the focal article. Even with this limitation this process generated 478,514 pairs to code. Our research assistants coded the article pairs in rank order until they had encountered five pairs in a row with no pairs having the

[20] We calculate each article's word counts after translating non-English articles into English using Google Translate.

[21] The Universal Sentence Encoder (Cer et al., 2018) is trained for transfer learning on NLP tasks. As of July 2024 the transformer-based version of USE ranks highly on the Semantic Textual Similarity benchmark dataset `https://paperswithcode.com/sota/semantic-textual-similarity-on-sts-benchmark` (Accessed July 28th, 2024). We use the deep averaging network (DAN) version of the USE for its additional gains in efficiency. Cer et al. (2018) find marginal gains for the transform version over the DAN version, although USE-DAN still performs better on a variety of a downstream transfer learning tasks than Word2Vec and CNN benchmarks.

[22] We randomized this for all but the first twenty focal articles we generated candidates for. For these articles we used both approaches to compare the performance of the ranking algorithms.

same underlying subject (persons, places, things, or events the articles were primarily focused on). We used this looser stopping rule rather than asking research assistants to stop when they had found five pairs in a row with no narrative commonality because our ranking approaches are targeted for semantic similarity rather than narrative commonality.

In this coding process our research assistants coded 1,631 unique pairs, identifying 86 pairs which made the same claim about the same event. These articles make up the set we use for our recall estimation.[23]

For our recall analysis we estimate the percent of these known positive cases our LLM-based approach versus alternative approaches could identify. For our LLM-based approach, we estimate the recall of three different aspects of our workflow. First, we estimate the recall performance of both steps of our SBERT candidate generation step to document how many positive cases in our recall set we lose prior to LLM annotation. Second, we pass the full recall set directly through our same claim, same subject GPT4o annotators. This estimate is to see how well GPT4o works as an annotator without having to worry about positive pairs dropped through the candidate generation process. Finally, we estimate the overall recall for our full SBERT-LLM approach. We divide how many positive cases are returned at the end of our method by the total number of positive cases in our recall set. For our alternative text reuse measure, we estimate the percent of recall pairs that were returned by a series of cosine cutoffs (.2, .4, .6) - e.g. what percent of known positive cases had at least .2 5-word gram cosine similarity? For our topic modelling based approach, we estimate what percent of known positive cases were placed in the same topic cluster.

Compared with recall, estimating precision is much simpler, as the denominator comprises all *positively labelled* data, $\frac{TP}{TP+FP}$. For each of our estimators we randomly sampled pairs of articles which were predicted to be positive. Research assistants evaluated these labelled positive cases for whether they were true "same claim, same event" pairs.[24] For our LLM-based method, we found

[23]With the exception of the ranked sets for the first twenty sets of focal articles, each set was coded by one research assistant. We validated the recall set by having each identified positive case be coded by a second research assistant. In cases where the two labels diverged one of the study authors reconciled the two labels.

[24]Research assistants exhibited slightly higher intercoder reliability when coding predicted positive pairs for the ngram measure (78.9%) and STM measure (80%) versus SBERT-LLM (70.4%). This discrepancy is likely because of the ngram and STM pairs were more likely to be uniform (uniformly positive cases for ngram, negative cases for STM)

that it was prohibitively expensive to directly estimate the precision of our LLM annotator due to the number of cases that would need to be coded to generate a sufficiently large sample of positive cases. We thereby restricted our precision analysis for this method to estimating the precision of our SBERT-LLM approach, rather than . We had research assistants label pairs that were identified as candidate pairs in the SBERT step and then were positively labelled as "same claim, same event" in the final LLM-instruct step. For the exact text approach, we had research assistants labeled pairs of articles with between 5-gram cosine similarity in a range of cutoffs (.2 to .4, .4 to .6, and greater than .6). We calculate the overall precision for each cutoff (.2, .4, .6) by estimating a mean weighted by the distribution of pairs in each bucket. For the topic analysis we had research assistants evaluate articles placed into the same topic cluster.

5.1 Recall Results

Overall for recall we find that our LLM-based measure performs much better than exact text measures, but slightly worse than our topic model measure. Furthermore, our SBERT-based candidate generation step successfully reduces the number of pairwise comparisons we need to make (and thus the overall computational and financial cost of the LLM-based measure) without sacrificing recall. In this section we first discuss the performance of our candidate step and present the recall rates for the different measures.

The following pair of plots demonstrates how our candidate generation process recalls almost all of the 86 recall set pairs while discarding more than 95% of potential pairs, or 5.6 million out of approximately 6 million potential pairs. First, in Figure 1 we display recall set results for the Bi-Encoder step, where we calculate embeddings for each article's English summary in our bioweapons case study and calculate the cosine similarity between embeddings. We show the between article cosine similarity for the approximately 6 million pairs not part of the recall set in red and the same distribution for the 86 pairs part of the recall set in blue. The dashed line shows the cutoff we set for

and thus easier to label. For this reason we required all SBERT-LLM pairs to be coded by three research assistants, with majority vote as final label. For STM and ngram labels we had pairs coded by two research assistants, with any discrepancies checked by one of the authors.

the first step of our candidate generation: .7 cosine similarity. Setting this threshold recalls 83/86 (96.5%) of the recall set while discarding 5.6 million or 93.6% of total possible pairs.

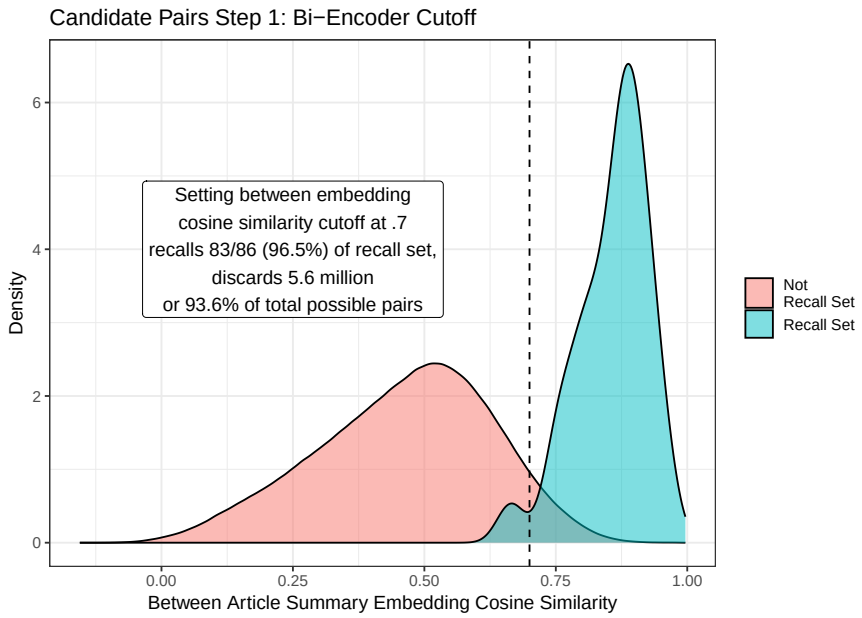

Figure 1: Bi-Encoder step discards majority of article pairs while recalling 96.5% of recall set. X-axis displays distribution of between article cosine similarity, measured on article summary SBERT embedding vectors. Potential article pairs not part of recall set (over 6 million pairs) displayed in red, while potential article pairs part of recall set (86 pairs) displayed in blue.

In Figure 2 we display recall set results for the Cross-Encoder step, where we compute cross encoder scores for the 392,320 pairs which passed the first stage Bi-Encoder step. Using the Cross-Encoder step allows us to further refine the number of candidate pairs while still recalling the vast majority of known positive cases in our recall set. Setting a cross-encoder cutoff at .5 still returns 83 out of the 86 recall pairs (96.5%, losing no additional known positives retained in the bi-encoder step). We are left with, however, only 64,677 candidate pairs to label with our LLM annotator. *Our candidate generation process thus recalls 96.5% of our known positive cases while discarding 98.9% of the 6,091,795 possible pairs in the bioweapons case study*.

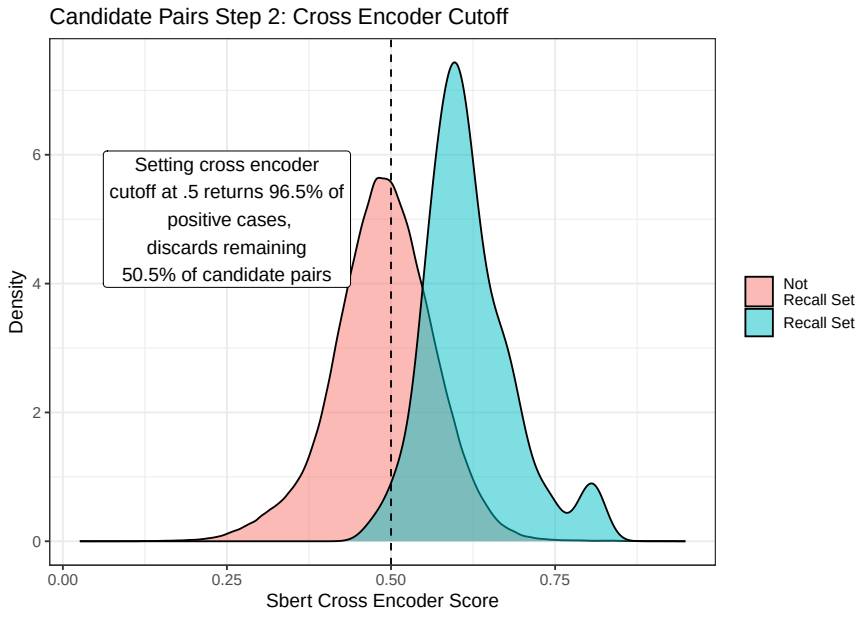

Figure 2: Cross-Encoder step reduces the number of candidate pairs to 64,677 while only discarding 9 additional known positive cases. X-axis displays distribution of between article cross encoder scores. Potential article pairs not part of recall set (X pairs) displayed in red, while potential article pairs part of recall set (83 pairs after bi-encoder step) displayed in blue.

The following table displays the recall rate for the exact text, topic modelling, and LLM-based measures. First, we provide the recall rate for our exact text reuse measure for a range of cutoffs (.2, .4, and .6). Second, we provide the recall rate for the topic clustering measure. Third, we provide the recall rate for our LLM. And finally we provide the recall rate for our SBERT-LLM approach, combining our candidate generation step with the LLM.

Estimator	Recall
.2 5-word Cosine Similarity	6.98%
.4 5-word Cosine Similarity	5.8%
.6 5-word Cosine Similarity	5.8%
Topic Model Clustering	95.3%
LLM only	77.9%
SBERT-LLM	73.3%

Table 2: Recall Rates for Narrative Commonality Measures

Overall we find that using an exact text reuse approach, even with a relatively low similarity threshold which only captures segments of overlap (.2 5-word gram cosine similarity), almost entirely

fails to recover our known cases of narrative commonality. This approach only recovered 6 out of 86 pairs in our recall set.[25] This is striking given that the majority of our recall set examples (50, or 58.1%) were identified through the ranked bag of words algorithm. We were thus stacking the deck in favor of an approach relying on exact overlapping textual features.

By contrast, our topic modelling and LLM-based measures recovered almost all cases. The topic modelling-based measure recalled 82 out of 86 cases (95.3%). The LLM annotator and SBERT-LLM approach recovered 67 and 63 cases, respectively, or approximately 73.3 to 79.9% of cases. For SBERT-LLM, we observed no drop off in recall with the candidate generation step: almost all cases that were recalled by our direct annotator were also recalled in the candidate generation process. These results provide motivation for our intervention and future methodological work on this topic: the dynamics of the media field, especially copyright law and journalistic norms, mean that exact copying is relatively rare outside of direct syndication agreements. Approaches which do not rely on exact text reuse represent a significant improvements over exact text approaches to identifying narrative and informational reuse in the commercial media environment.[26]

5.2 Precision

One of the reasons journalism and media scholars have relied so heavily on exact text reuse to identify information and narrative diffusion is because these approaches have very high precision: if two articles have very long overlapping sequences of texts it is unlikely they were independently generated. We find warrant for this position here, as despite the low recall performance of exact text reuse approaches vis-a-vis our SBERT-LLM approach, they nonetheless out-perform on precision. On the flip side, our topic model approach greatly underperforms vis-a-vis SBERT-LLM. The following table shows the percent of predicted positive cases which research assistants also labeled

[25] We also tried using a smaller context window for measuring reuse - tri-word gram cosine similarity. We found no differences in our results. Only 8 cases were recalled with this more flexible context window.

[26] One thing we do not test here is the performance of a bag-of-words estimator against our LLM annotator. It would be circular to do so, as our bag-of-words ranked annotation process ensures that pairs which have high between article word cosine similarity were more likely to be identified in our recall set.

as "same claim, same subject."[27] As noted in the section introduction we only consider precision of the overall SBERT-LLM approach.

Estimator	Precision
.2 5-word Cosine Similarity	66.9%
.4 5-word Cosine Similarity	79.4%
.6 5-word Cosine Similarity	100%
Topic model (STM)	15.6%
SBERT-LLM	34.1%
SBERT-LLM 3-day	42.9%

Table 3: Precision Rates for Narrative Commonality Measures

We see that our SBERT-LLM approach performs substantially better in terms of precision than the topic modelling setup, but substantially worse than our exact text estimators. In looking at the individual mislabelled cases, part of the challenge is that a single story may evolve over several days and episodes, with the same individuals or organizations repeating the same claims in different venues. For example, one mislabelled case was a pair of articles referencing Joe Biden's attendance of a NATO summit. Both articles made similar statements and claims related to a hypothetical U.S. response if Russia were to use nuclear and other weapons. One article, however, focused on the context of Biden's NATO appearance before the summit, and one article detailed statements Biden made after the summit. In this case as in others it is true that the pair of articles have similar subjects and claims, but a human annotator would recognize the two articles as focused on separate days and events.

This case and others suggests one way to improve our LLM annotator estimator is to restrict our matching process by day. Indeed we find that if we restrict our estimator to matching within a three day window, our overall precision increases (in the precision sample) to 42.9%. We also don't lose much on recall because 80.2% of our recall pairs were published in a three day window. For this

[27] In total we had research assistants label 100 pairs predicted by our SBERT-LLM approach to be "same claim, same subject" and thirty pairs for each of the ngram cutoffs. We found positive cases from SBERT-LLM to be more challenging for the research assistants to code (agreement between two coders ranged from 69.4% to 76%), as they were more likely to be edge cases (due to the candidate selection process). For this reason we had each pair coded by three humans and took the majority vote across the three labels. For the ngram analysis pairs of research assistants agreed on average 75% of the time. For this analysis the majority of pairs were only coded by one human coder, with 22% coded by two. Any discrepancies in the overlapping pairs were voted on by a third labeller.

reason in the following case study we use this narrower date window.

5.3 F1

One way to quantify a precision-recall tradeoff between methods is to compute F1, which is the harmonic mean between precision and recall. F1 is defined as $F1 = 2*(precision*recall)/(precision+recall)$.[28] Results for all measures follow:

	Recall	Precision	F1
0.2 5-word cosine similarity	6.98	66.9	12.6
0.4 5-word cosine similarity	5.8	79.4	10.8
0.6 5-word cosine similarity	5.8	100.0	11.0
Topic Model Clustering	95.3	15.6	26.8
SBERT-LLM	73.3	34.1	46.5

Table 4: Precision, Recall, and F1 metrics (multiplied by 100) for Text Estimators and SBERT-LLM approach.

What's clear based on Table 4 is that our the LLM approach represents a substantial improvement over the exact text matching and topic modelling (STM) estimators we tested.

6 Application: Diffusion of Russian State Media Narratives

We now turn to our application—understanding how the false claims that Ukraine and the United States were developing biological weapons for use against Russia, which appeared in Russian state media, were covered in the U.S. media outlets in our corpus. We first outline our hypotheses concerning which types of media outlets were more likely to reprint Russian claims, and then detail our results.

[28]The F1 score is the harmonic mean between precision and recall. One note is that our precision and recall estimates are not drawn from the same population. Precision is $\frac{TP}{TP+FP}$. Recall is $\frac{TP}{TP+FN}$, where 'TP' is true positive, 'FP' is false positive, and 'FN' is false negative. The challenge in our setting is that the set of true positives is not the same for the recall and precision calculations, as our recall estimation only focused on a subset of all the bioweapons articles. We nonetheless include this estimate in order to capture this tradeoff between precision and recall.

6.1 Hypotheses

Low quality news We hypothesized that low quality news outlets with lower journalistic standards would be more likely to print narratives appearing in Russian state media outlets (Miskimmon and O'loughlin, 2017; Oates and Ramsay, 2024; Ramsay and Robertshaw, 2018; Oates et al., 2020; Watanabe, 2017). There have long been pressures on digital-native news outlets to cut costs and lower journalistic standards, with an increasing emphasis on audience engagement metrics (Mothes et al., 2024). At the same time, "alternative media" outlets that do not adhere to conventional journalistic aims and norms, and which may stimulate demand for sensational rather than factual content have become evermore prominent (Strömbäck, 2023). While this may be linked to the digital age, many have drawn comparisons between the lack of journalistic standards in alternative media and past media systems in place before modern U.S. journalistic standards, which were similarly characterized by different aims (this era was characterized by partisan news outlets, and often "yellow journalism" Mutz and Young, 2011). Thus, these alternative media outlets with a high premium on sensationalism may be especially likely to print outlandish claims espoused by Russian State media, and to do so without debunking them.

Responsiveness to newsworthy events We also hypothesized that newsworthy, real world events covered by Russian State media would also be more likely to appear in U.S. news outlets. This hypothesis reflects the simple fact that state media outlets necessarily cover ordinary news, and do not exclusively produce content designed to persuade audiences. As media scholars have posited, most media report on emerging crisis events reactively (Boydstun, 2013; Zaller, 2003), and do so because they cover what audiences believe is important (Iyengar and Kinder, 1987). Classic work in this field has found that the power of the media to set the national agenda is strongly limited by the salience of "real world events," which simultaneously affect media coverage and public sentiment (Iyengar and Kinder, 1987).

6.2 Results

Our results demonstrate that low quality US news websites were much more likely than mainstream US news websites to publish stories related to the bioweapons that had overlapping claims and subjects with Russian state media but not Ukrainian news sources. Low quality US news websites in our database printed 369 articles related to the bioweapons case study. Of these, 69 (18.7%) shared information with Russian but not Ukrainian sources, 7 (1.7%) shared narratives with Ukrainian but not Russian sources, and 31 (8.4%) shared information with both. Of the 683 bioweapons related articles published on mainstream US news websites, 20 (2.9%) shared information with Russian but not Ukrainian, 77 (11.7%) shared information with Ukrainian but not Russian sources, and 51 (7.5%) shared information with both.

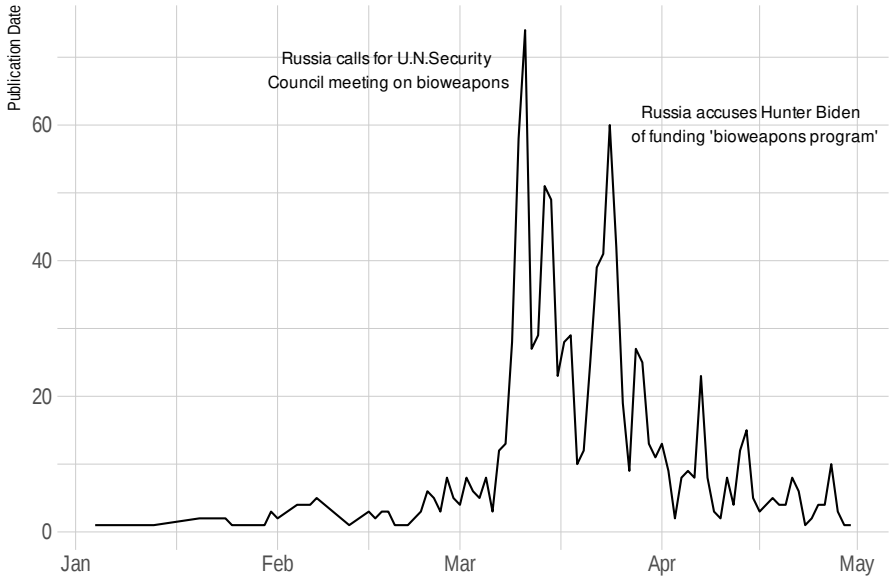

Figure 3: U.S. media articles matching Russian State Media stories about bioweapons. Spikes in media coverage correspond to March 11, when Russia called for U.N. Security Council meetings to discuss its accusations about Ukranian bioweapons, and March 24, when Russia launched accused President Biden's son Hunter of securing funding for the "bioweapons program."

The following plot shows the distribution over low quality (left) and popular U.S. media sources (right) for the percent of articles which shared narratives shared with Russian versus Ukranian.

We display both the individual data points as well as a box plot estimating the overall univariate distribution.

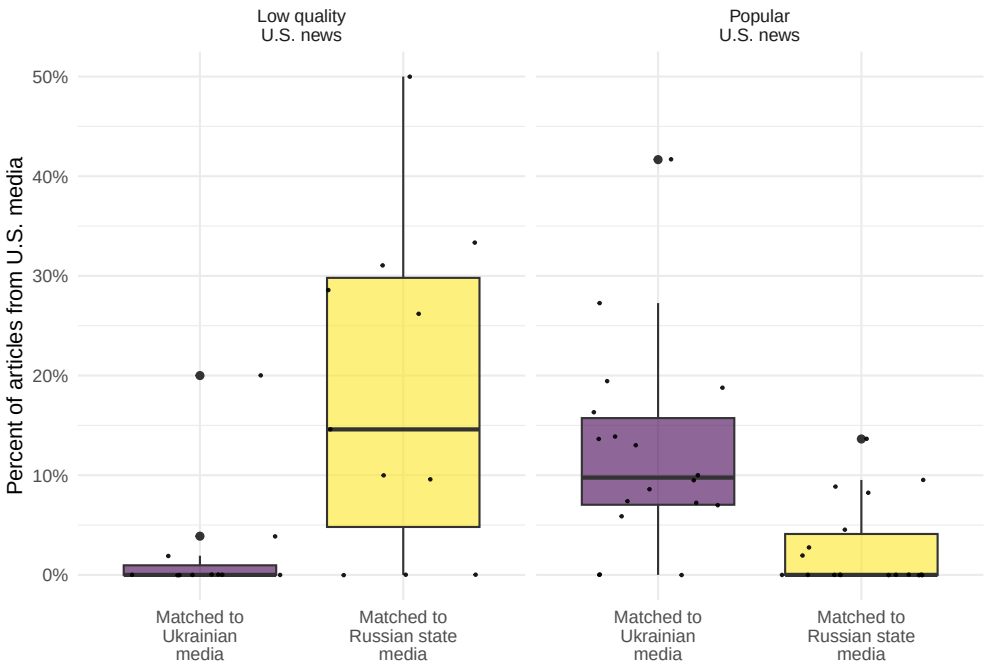

Figure 4: Percent of U.S. media articles that share narratives with Russian state media and Ukranian media outlets on bioweapons. Low quality U.S. media websites (left) are more likely than mainstream popular US news sources (right) to print stories that contain the same narratives as Russian State media articles.

These findings establish that low quality news sources played a role in circulation of information also highlighted in the Russian press in the case of various narratives around Ukrainian bioweapons. However, these findings do necessarily mean that low quality U.S. news outlets reprinted Russian propaganda—it's possible that the claims originated in low quality media before circulating to Russian state media or in a third, unobserved source.

We also investigated which topics were shared between U.S. news sources and the Russian state versus Ukrainian media outlets, among stories mentioning bioweapons. The following plot conditions the percent of articles which had the same narrative as a Russian state media article (left) and the percent of articles which had the same narrative as a Ukranian media outlet (right) by topic and type of source (most popular US news vs. low quality news). We group articles by topic

by setting a threshold for topic prevalence. Articles which have greater than the topic prevalence threshold we thus consider "on that topic."[29]

[29]We estimate document topics using the topic model introduced in the validation section above. See Supplemental Index Section D for more details. We set a threshold of .2 topic prevalence because above that threshold the number of documents unassigned to any topic (i.e. with no topics meeting that threshold) increases precipitously. Substantantively this threshold can be interpreted as indicating that at least 20% of a document's words were estimated to have been drawn from that topic's distribution over the corpus vocabulary. Setting a threshold at .2 means some documents will be repeated in the figure, meaning they have more than one topic above that threshold. Of the 3,351 documents that were assigned to at least one topic, 2076 articles were assigned to one topic, 1,167 articles were assigned to two topics, and 108 articles were assigned to three topics. We treat these topics as observed and estimate standard errors based on one-sample proportion standard errors. These standard errors are likely artificially deflated due to the non-i.i.d. nature of clustered data as well as error propagation from our "same claim, same subject" classifiers (**?**).

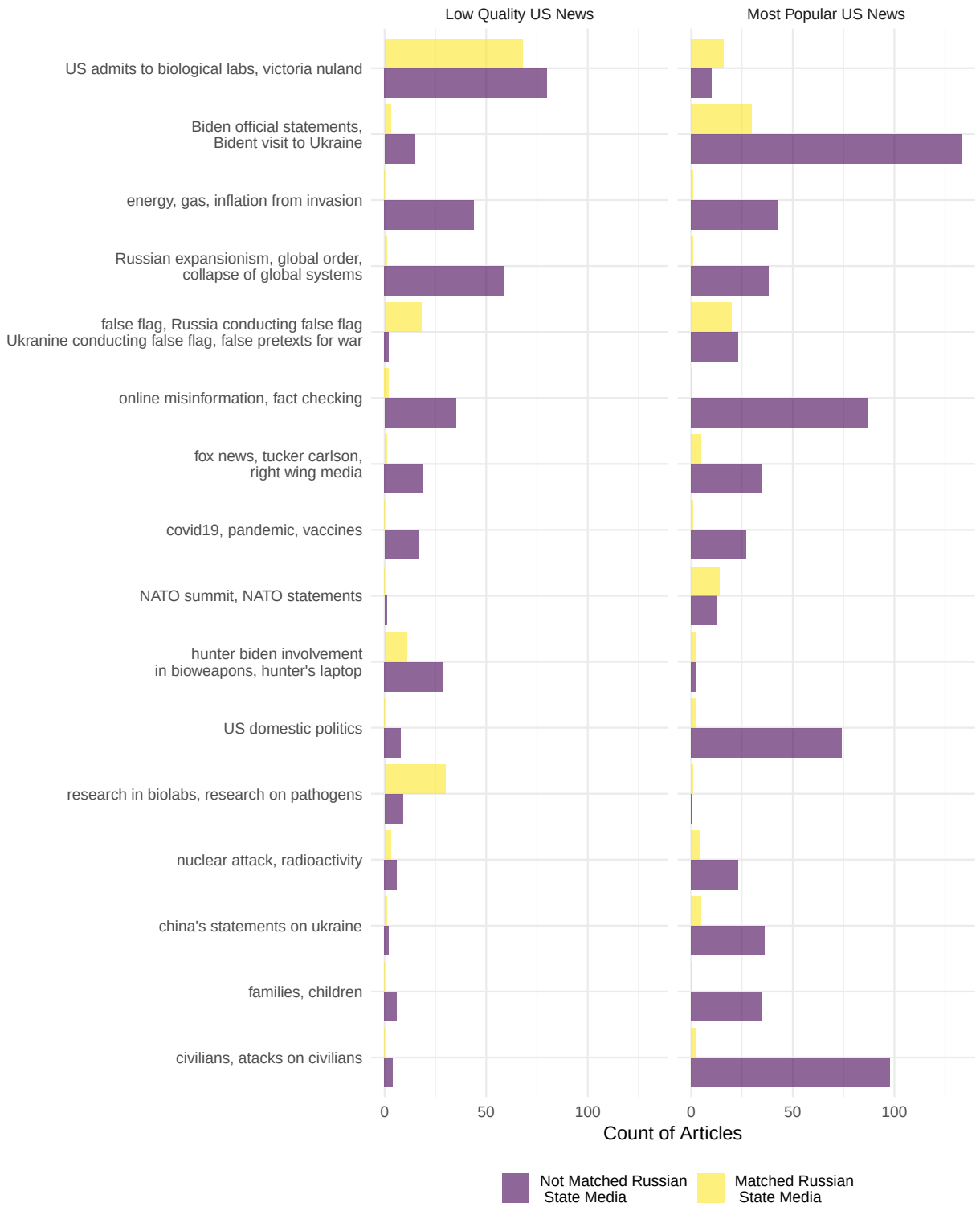

Figure 5: Percent of Low Quality and Popular News Articles sharing narratives with Russian state media (left) and Ukranian media (right) by topic. Articles were assigned to topics if their estimated prevalence for a given topic was greater than .2 or 20%. Topics limited to those with at 20 articles in the US news (low quality or most popular) corpus. See Supplemental Index, Section D for the full figure, including topics with low numbers of documents.

We find that when we condition on topic, we find that low quality and most popular news outlets largely track one another with a few key exceptions. First, we see a higher rate of commonality between mainstream US news and both Russian and Ukranian sources when reporting on NATO-related events. This finding is more consistent with the likelihood that for certain fact-based events mainstream U.S. news sources, Russian state media, and Ukrainian media were all reporting on the same sequences of events with the same underlying claims or narratives.

Second, and most strikingly, for discussions of false flag operations, the idea that Russia (or Ukraine) would a stage an attack, blame it on the other, and use it as a false pretext for further conflict, we observe that low quality news coverage was more likely to have narrative commonality with Russian state media articles than mainstream U.S. news sources. This is flipped when we look at narrative commonality with Ukranian sources. In this case mainstream U.S. news sources were much more likely to have narrative commonality than low quality sources. Without examining this case longitudinally we cannot know for sure, but this pattern is suggestive of low quality news circulating talking points that emerged in Russia.[30]

To demonstrate how this approach can be used to trace specific narratives over time, we dig into the case of Russian state accusations that the U.S. was planning a false flag operation. Importantly, we identified these articles as a connected component of the graph formed by our pairwise comparisons indicating a common narrative (not using the topic model data, which included a number of other articles). Figure 5 shows that this narrative was first covered in the Russian state media outlet Sputnik, then in Ukrainian media (warning the public about it), and finally in U.S. low quality outlet Infowars.

[30]See the Supplemental Index, Section D for topic definitions.

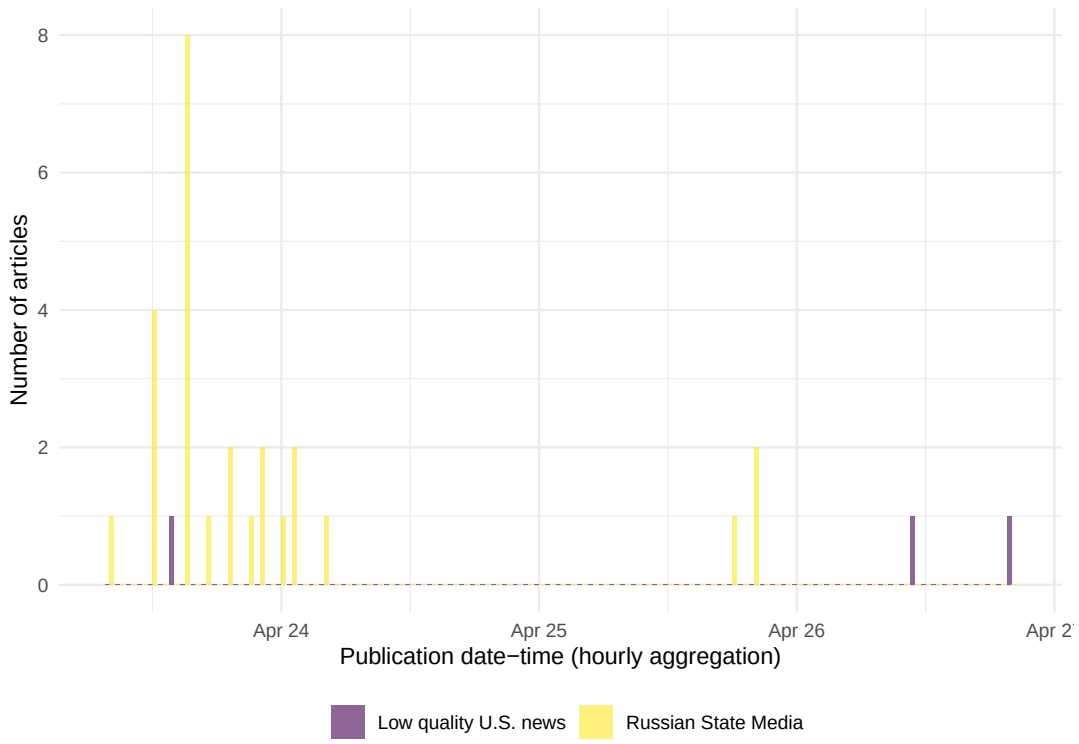

Figure 6: Total articles published on Russian accusations of a U.S. "false flag" operation that would make it appear that Russians used WMD on the battlefield, by article source.

We point to three notes of caution surrounding this approach. First, low precision for our narrative commonality estimator means that there are likely a number of false positives in these data. Second, we cannot know the direction of influence in these stories due to the likelihood of unobserved articles. Finally, we have selected for our case study a set of stories where we know Russia was producing a lot of content in a bid to influence western media coverage. The power of our approach is its scalability, and in future work we will expand this analysis to include a much broader set of stories (for example, all media coverage of the Ukraine war).

7 Conclusion

We have proposed "narrative diffusion," a phenomenon with the observable implication that two writers make the same specific claim about the same specific event—particularly those producing journalism. Social scientists and sociologists in particular have long sought to study the processes

of diffusion that characterize the media ecosystem—an extremely tempting prospect because of the sheer volume of potential data freely available and especially so in recent years with data and computational infrastructure so abundant. Doing so would allow us to shed light on phenomena as wide ranging as how modern journalistic practices involving borrowing from each other, how new ideas rise to prominence in our culture, and how academic fields borrow from each other. Yet our ability to gather data that synthesizes the claims documents make and compares them has proven to be an impossible task without the advances in language modeling in the early 2020s.

Thankfully, we are now able to gather data on common narratives across documents and study narrative diffusion. We have proposed doing so by identifying these pairs, we propose a three stage classification strategy: first, we distill source documents to their key claims using chain of thought prompting; then we identify candidates texts using a set of large model embeddings; and finally, we estimate whether these texts are indeed making the same claim about the same event based on a series of progressively costly large language model classifiers to first identify candidate matches (often edge cases) and then prompt an instruction-tuned large language model based on chain-of-thought. We validate this classifier using an innovative approach to estimate conventional classifier metrics of interest involving identifying edge cases—and show our approach identifies orders of magnitude more cases of narrative diffusion compared with oft-used exact text matching approaches (e.g., Bail, 2012, 2015; Hinkle, 2015a; Wetts, 2023).

We apply our method here to a particularly important case—U.S. news coverage of narratives from Russian state media outlets surrounding the false accusation that Ukraine and the United States were developing bioweapons for use against Russia. We find that even on this contentious issue, cases in which U.S. and Russian state media sources print the same narratives often reflect ordinary coverage of real world events. However, we do find that a great deal of coverage in low quality U.S. news sources does indeed share common narratives with Russian propaganda. For example, we find that low quality U.S. sources Infowars and Natural News printed stories covering Russian accusations that the US was planning a false-flag operation that would make it look as if Russia used weapons of mass destruction on the battlefield.

There are a number of limitations inherent in this approach and in this paper specifically. The data collected here are subject to the classic inference problem in science: simply because two things happen close in time does not necessarily mean there is a causal link, even when we observe temporal precedence. More specific to our paper is that this approach might be "overfit" to documents related to the Ukraine war, though we suspect tweaks to the input data and prompt text may address this. We also do not evaluate this approach against those that use advanced text features from the traditional NLP literature (e.g., Ash et al., 2024; Wang and Rambow, 2024). There are other more mundane limitations as well—due to non-standard protocols across media websites, metadata and date data specifically may be especially error prone in the data we've collected.

This application represents an initial effort and there is a great deal to do both in terms of methodological development and analysis. Methodologically, future work should seek to improve the precision of the estimator (i.e., aim for fewer false positives) and seek methods to improve the accuracy of "candidate matches," as instruction-tuned LLMs are expensive to use and deploy. Unfortunately our experience working with open-weight LLMs such as Llama 2 and Llama 3 suggest that open-source deployments are *not* likely to result in cost savings any time soon given the number of annotations pair-wise comparisons require. Future work should also seek better candidate generation approaches to allow for a scaling factor greater than 100x, and investigate methods to scale this up more broadly.

There are interesting methodological problems to address related to this approach beyond simply improving the performance of LLMs. For example, because our LLM approach (at least seems) less accurate than human coders, one ought to apply debiasing approaches from the surrogacy literature when computing statistical quantities of interest that depend on this data (Egami et al., 2023). What's more, that literature has implications for estimating the variance of said statistics—networks are not independent and identically distributed (IID), thus one must account for network correlation to avoid generating anti-conservative estimates of variance. Because the network that characterizes the dependence in the data generated by our approach is estimated using the same imperfect surrogates, one also needs to *debias estimates of the variance*, which to our knowledge

is an unsolved problem.

Despite these challenges, we argue that large language models offer a unique opportunity to measure quantities of interest that are difficult to measure with existing approaches.

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Supplementary Materials ***for***

Quantifying Narrative Reuse Across Languages

Contents

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A Data Availability Statement

We are committed to sharing a full repository at the time of publication with all code and intermediate data files necessary to reproduce the analysis in this article. Due to intellectual property law we will not be able to share raw text.

B Data Collection

We created a dataset of news articles through an extensive process of collecting, downloading, parsing, and cleaning the full text of news articles from an array of different news outlets. We began by defining a list of media outlets, including both high- and low-quality US news and both Russian and Ukrainian news sources. We then collected the urls for all articles published by these sources (and in some cases, their Russian and Ukrainian language counterparts) between January

1st, 2022 and April 30th, 2022. For each article, we scraped the raw html and used our custom parsing templates to extract structured text fields from each document. Lastly, we ran multiple stages of validation to ascertain the quality of our dataset, measuring both the completeness of our set of articles and the precision of our text parsing. This section first provides additional details on how we selected our sources, then discusses how we collected and parsed the html for individuals articles, and finally provides a validation of the coverage of our sources.

B.1 Selection of sources

Our sources include news content from US popular mainstream news sources, low quality US news websites, Ukranian news websites, and Russian state media. We first started with a larger set of sources, identifying the twenty-five most popular news sources, twenty US low quality sources, four state owned Russia state media outlets, and seventeen high quality Ukrainian news websites.

For each site we attempted to scrap all English, Russian, and Ukranian language variants. For increased specificity, we provide details in these supplementary materials for each individual language variation of a source. Language variations for a given source are differentiated by appending the language code to the end of the source name (e.g., ukrinform_uk and ukrinform_ru refer to the Ukrainian and Russian language versions of Ukrinform respectively).

Our final data set is a subset of these sixty-six sources. We eliminated sources which we were unable to collect because content was paywalled (Wall Street Journal, LATimes) or the site had become defunct over our time period (wnd.com, collective-evolution.com). We also eliminated sources with poor coverage over our study period.

B.2 Collection of article urls

We began collecting articles at the beginning of March 2022 from the news websites we identified. At this time, we set up RSS feed scrapers to collect links for 16 sources and added additional sources in the following weeks. The RSS feeds we used were produced by the news sources themselves and provide real-time feeds of published articles. The benefit of this approach is that the source of articles is credible and it allows for real-time collection. But a limitation to this data source type is that it is not historical and consequently not suitable for retrieving older articles. To address this, we collected articles from the historical archives maintained by individual websites.

One challenge we encountered with including URLs selected from website archives is that it is challenging in some cases to distinguish what counts as a "news article." RSS feeds present articles in a finely structured way such that we can parse the article metadata very easily and have confidence that each item is actually an article (instead of a link to some various page on the site).

However, news website archives sometimes present articles in an unstructured way or include links to pages other than articles. In these cases, we were required to identify what counts as article urls.

We used a liberal collection method to ensure we captured all article urls at the cost of also collecting non-article urls. We then used a multi-stage filtering method to exclude urls for pages that are not articles. This method included removing urls from sections known to be not articles (i.e., author bio pages) and removing urls that failed to match any of our parsing templates (if the html of a page matches the expected format of an article page, we include it as an article; otherwise, we exclude it).

B.3 Collection of article HTML

After identifying our set of links for each source, we downloaded the raw HTML for each page. For both RSS-collected articles and articles collected via a source's web archive, we ran into the challenge that if there was a delay between link collection and HTML download it's possible that we could generate downstream source bias in our analysis. This is because it's possible that news websites edited their articles post-publication.

To correct for this problem, we download the raw HTML of articles from Wayback Archive snapshots for any article whose url was collected not in real-time. The Wayback Archvie collects snapshots of web urls on a regular basis, sometimes multiple snapshots a day. We also used Wayback Archive snapshots for articles that we were not able to scrape, such as those that were blocked by Cloudflare. For all other articles (i.e. those collected in real-time from an RSS feed), we download the raw html of the article from the original collected link from the source. We deployed Python web scrapers to download the raw HTML for articles, using a combination of the built-in Requests library and Selenium, a more advanced web scraping tool that automates a browser session and was necessary for some of the Russian and low-quality US sources since they block access for basic web scrapers. Full details on where page HTML was downloaded from and the download delay between article publication and either the timestamp of when we downloaded it or the Wayback Archive snapshot are included in A1.

The following table details for each source the count of articles that we immediately collected raw html for (i.e. downloaded within one day) versus the count of articles there was a longer delay between publication html capture. It also details whether a given website's HTML was downloaded from the internet archive only, source only, or both.

Source	**Downloaded within 1 day**	**Downloaded within 1 week**	**Downloaded greater than 1 week**	**Avg download delay (days)**	**Download source**
100_percent_fed_up	762	105	68	235	both
abc_news	6912	42	222	20	both

Table A1 continued from previous page

Source	Downloaded within 1 day	Downloaded within 1 week	Downloaded greater than 1 week	Avg download delay (days)	Download source
bipartisan_report	428	535	48	202	both
business_insider	10468	268	31	24	internet_archive_only
bykvu_eng	47	180	470	179	both
bykvu_ru	2	0	5907	295	both
bykvu_uk	5403	274	224	91	both
cbs_news	5383	41	10	75	both
censor_net_en	884	630	251	50	internet_archive_only
censor_net_ru	4741	3807	3389	40	internet_archive_only
censor_net_uk	10455	3911	2839	31	internet_archive_only
clash_daily	550	52	0	0	internet_archive_only
cnbc	8560	89	59	240	both
cnn	10458	338	168	47	both
daily_caller	7049	23	3	33	both
fakty_ru	227	65	6060	230	both
fakty_uk	258	168	5925	216	both
fox_news	18556	1077	152	176	both
gordonua_en	4	3	9	42	internet_archive_only
gordonua_ru	16081	167	186	195	both
gordonua_uk	1785	346	3216	32	both
huffpost	6459	8	0	0	both
ijr	1779	225	98	84	both
infowars	3208	13	0	0	internet_archive_only
interfax_ua_en	3955	3	1043	80	internet_archive_only
interfax_ua_ru	8598	795	4017	157	both
interfax_ua_uk	9903	35	2900	101	both
lb_ua_en	501	127	107	124	both
lb_ua_uk	12178	297	47	69	both
msnbc	3981	424	640	227	both
natural_news	3127	222	14	22	both
nbc_news	10205	970	77	165	both
new_york_post	22466	189	31	129	both
new_york_times	16168	241	24	60	both
npr	0	0	7057	414	source_only
nv_ru	0	0	756	292	source_only
nv_uk	0	0	636	306	source_only
palmer_report	1043	25	2	176	both
pbs	1872	1	0	0	internet_archive_only
politico	1273	3	3	11	both
pravda	684	61	5	66	both

Table A1 continued from previous page

Source	Downloaded within 1 day	Downloaded within 1 week	Downloaded greater than 1 week	Avg download delay (days)	Download source
pravda_com_en	3266	115	48	116	both
pravda_com_ru	11189	148	292	246	both
pravda_com_uk	11204	61	287	160	both
pravda_ru	14684	273	283	168	both
rbc_ru	19617	158	156	134	both
rbc_uk	16754	769	1811	156	both
rt	7409	230	261	231	both
rt_ru	29129	1465	9337	113	both
slate	1918	73	65	108	both
sputnik	8847	206	66	130	both
stillness_in_the_storm	1741	568	1239	239	both
tass	5389	18	5790	375	both
tass_ru	38186	1994	41184	332	both
the_gateway_pundit	2321	32	8	49	both
the_hill	12075	87	139	275	both
the_political_insider	0	0	861	292	source_only
tsn_en	195	165	199	206	both
tsn_ru	23363	807	8275	276	both
tsn_uk	31188	941	1459	142	both
ukrinform_en	37	4	0	0	internet_archive_only
ukrinform_ru	129	46	87	159	both
ukrinform_uk	534	16	42	72	both
unian_ru	14396	231	6877	107	both
unian_uk	14918	297	6871	103	both
usa_today	8845	2294	10341	292	both
washington_post	3852	388	377	260	both
zn_ru	11896	370	115	130	both
zn_uk	10565	972	848	24	both

Table A1: When articles were downloaded and from where. Download delay values cumulatively sum to the number of articles in the period of analysis. The average download delay in days only refers to articles that were downloaded more than 1 week after the article publication date.

B.4 Parsing article HTML

Article parsing involves transforming the raw HTML of a web page into structured text suitable for downstream analysis. This process involved both manual and automated steps to first identify patterns in article HTML and then programmatically extract text fields based on those patterns.

Specifically, we created a set of parsing templates for every source where each template corresponds to a unique type of article HTML formatting. For example, a CNN article from the US Politics section has HTML formatted differently than an article from the Travel or Style sections and each would require a unique parsing template. Each individual parsing template is made up of a series of XPaths, which constitute specific patterns in HTML structure to identify a unique location or set of elements. We used XPaths to identify each text field of interest, such as the article author, title, and published date. For example, a parsing template will have a unique XPath pointing to the location of the HTML text element(s) containing the title of the article. In total, we created 156 templates which include over 1,000 XPaths across all our sources. Creating these parsing templates was a tedious, iterative process through which we identified each unique structure of HTML in our data and improved the precision of our templates. Once we created our templates, we parsed all the article raw HTML and ensured that each article matched one, and only one, unique template. The output was structured text data where for each article we had at least the article's author, title, published date, and the full text of the body of the article.

Parsing validation occurred in between the iterative steps of article parsing. After the parsing templates were initially created and the articles were parsed the first time, we conducted a validation exercise with the assistance of RA's to identify parsing errors which were then remediated before the final dataset was parsed. We sampled parsed articles, stratified by source and parsing template, and presented them to RA's with instructions for how to compare the parsed structured text from each article with the text from the original document. Specific errors were noted as well as overall problems with parsing templates. These errors were fixed and the parsing templates were updated. Finally, we used our refined parsing templates to generate the final dataset.

B.5 Validation of link collection

We performed a rigorous validation on the set of articles collected for our dataset to ensure that we have high confidence that articles are not missing. Our analysis of differences between high and low quality US news sources necessitates that we perform this validation. Otherwise, we might interpret some sources not sharing narratives with Russian state media as indicative of their true underlying behavior when in fact it was due to missing articles. This was a further motivation for why we generated our own dataset instead of relying on outside data vendors.

We validated our collection by comparing our project dataset with a validation dataset of articles collected via a different method. We created this validation dataset by extracting links from Wayback Archive snapshots of source subsections. Specifically, we identified all the subsections of each news site we wanted to validate, such as the Politics or International News sections of a site. Then we identified which days the Wayback Archive had available snapshots for all subsec-

tions for each news site. For each source, we sampled one day from each month of our period of analysis (January - April, 2022), downloaded the snapshots for each subsections, and extracted all links from each snapshot. Similar to our previous data collection strategy, we start with the set of all links to ensure we do not exclude articles at the cost of including links to non-articles. We then used the same process of parsing snapshot raw HTML outlined above to limit these links to only articles. We use this set of article links to evaluate the main dataset.

An overview of the validation dataset is seen in A2. We collected 152,126 articles across 49 sources for the validation dataset. Some sources from the main dataset are not included in this dataset due to an insufficient availability of Wayback Archive snapshots. In total, there are 9,868 articles missing across 34 sources, meaning we found no missing articles for 15 sources. Additionally, we provide the percentage of validation articles for a given source that were missing from the main dataset. When looking at these values, we see that 27 sources had a missing rate of less than 5% and the sources with the highest missing rates are concentrated around the Russian and Ukrainian language sources.

Source	**Validation Articles**	**Missing**	**Percent Missing**
abc_news	2663	17	1
bipartisan_report	582	162	28
bykvu_eng	145	0	0
bykvu_uk	302	0	0
cbs_news	2254	136	6
censor_net_en	705	74	10
censor_net_ru	3189	485	15
censor_net_uk	3245	490	15
clash_daily	476	0	0
cnbc	4515	849	19
cnn	4817	700	15
daily_caller	735	5	1
fakty_uk	29563	506	2
fox_news	2620	2	0
huffpost	19209	12	0
ijr	411	36	9
interfax_ua_uk	957	11	1
msnbc	3614	714	20
nbc_news	416	0	0
new_york_post	1694	124	7

Table A2 continued from previous page

Source	Validation Articles	Missing	Percent Missing
new_york_times	12162	105	1
npr	1051	131	12
nv_ru	2508	0	0
nv_uk	3599	484	13
palmer_report	203	0	0
politico	1054	101	10
pravda	472	1	0
pravda_com_en	159	0	0
pravda_com_ru	333	0	0
pravda_ru	2729	8	0
rbc_ru	376	0	0
rbc_uk	2681	0	0
rt	1204	196	16
rt_ru	1461	39	3
slate	1325	129	10
sputnik	1939	90	5
stillness_in_the_storm	1045	0	0
tass	398	58	15
tass_ru	5169	893	17
the_hill	10207	380	4
tsn_en	1539	0	0
tsn_ru	2418	506	21
ukrinform_en	482	0	0
ukrinform_ru	1108	616	56
ukrinform_uk	1301	900	69
unian_ru	4762	0	0
usa_today	3450	628	18
washington_post	2862	280	10
zn_uk	2017	0	0

Table A2: Articles in validation dataset and percent missing.

C Data Composition

Included below are a series of plots showing the daily count of articles for the sources in our data. We separate each plot by source type: low quality US, mainstream most popular US sources, Russian state media, and Ukranian media sources.

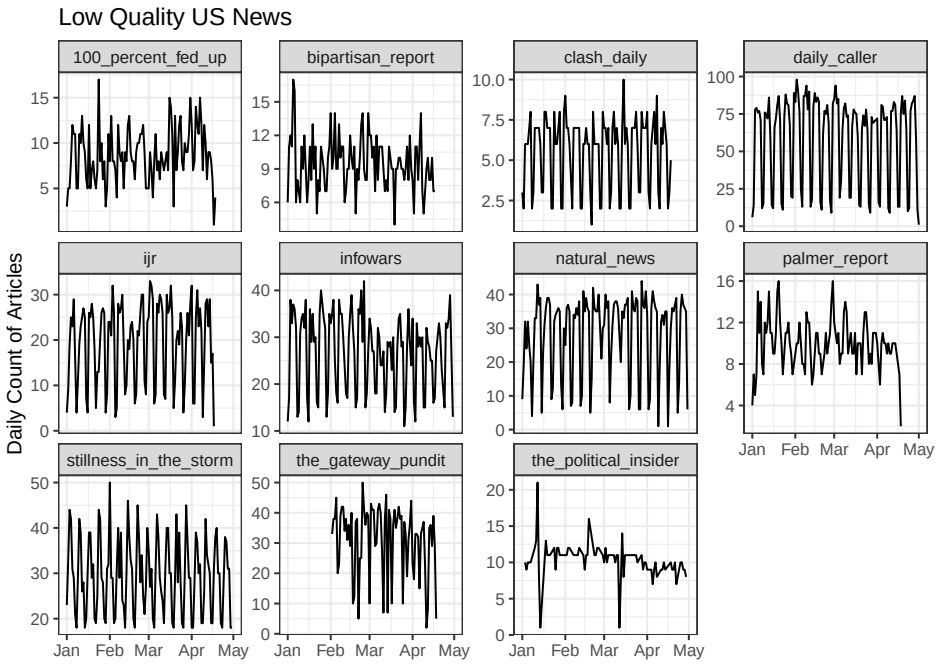

Figure A1: Article Counts by Source and Date for Low Quality US News Sources

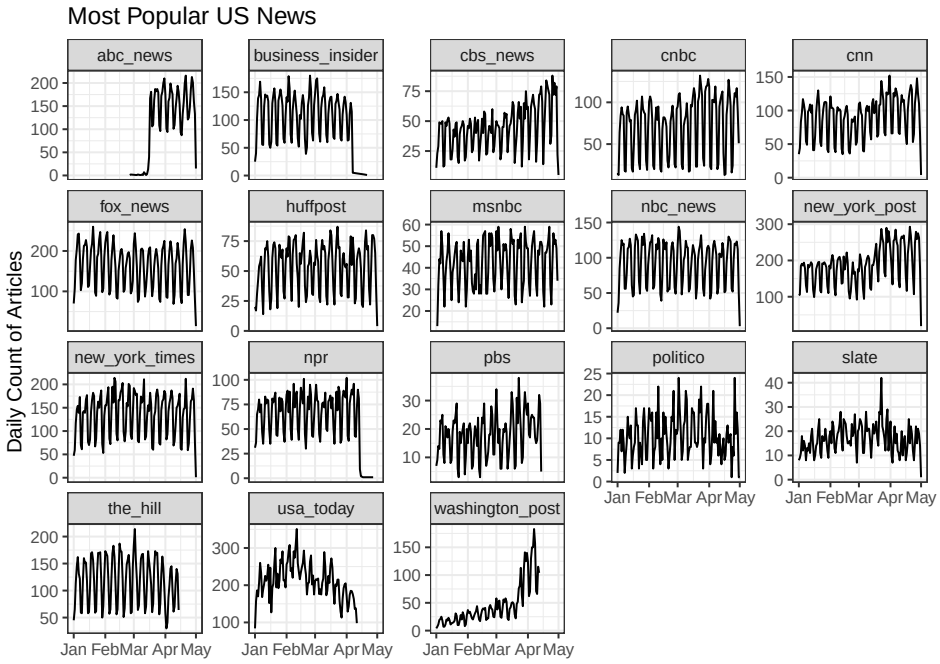

Figure A2: Article Counts by Source and Date for Popular Mainstream US News Sources

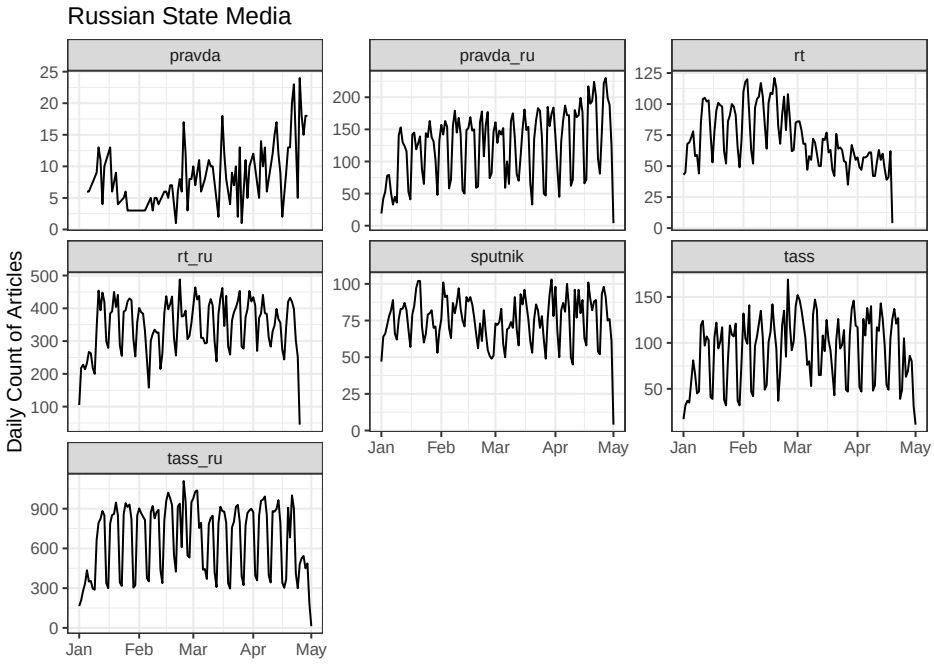

Figure A3: Article Counts by Source and Date for Russian State Media Sources

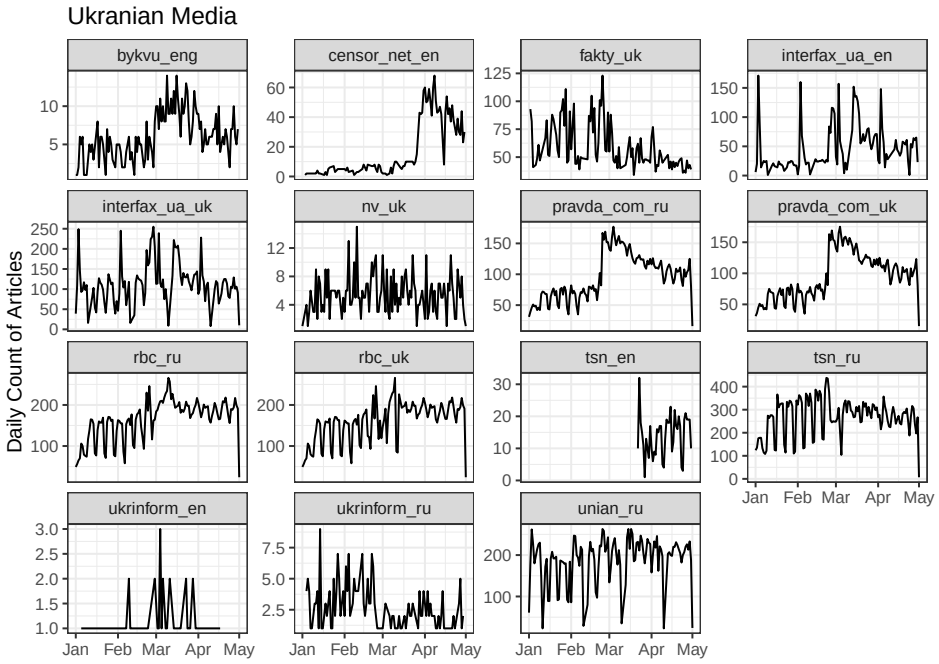

Figure A4: Article Counts by Source and Date for Ukrainian Media Sources

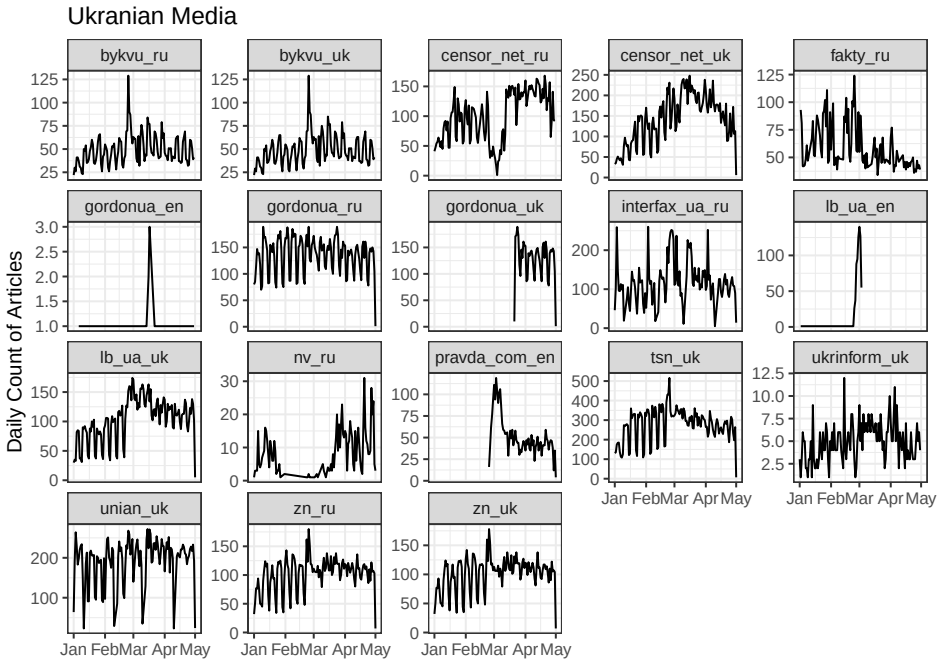

Figure A5: Article Counts by Source and Date for Ukrainian Media Sources

D Topic Model

This section discusses details of our topic model used as an alternative measure for narrative diffusion and as a covariate in our application. We first discuss the process of model fitting and then present some additional results.

D.1 Model Fitting

In order to understand to what degree topic models can map onto our estimand of narrative reuse, we fit a structural topic model (Roberts et al., 2019) to our bioweapons case study documents. We selected a topic model with thirty topics based on topic exclusivity, semantic coherence, and meaningfulness of the individual topics.

The following plot compares topic exclusivity and semantic coherence scores for a range of topic models (K = 20, 30, 40, 50, 60, 70, 80, 90, 110, 120). Semantic coherence measures the degree to which the most probable words co-occur in the same document (Mimno et al., 2011). Roberts et al. (2014) argues that semantic coherence should be balanced against topic exclusivity, the degree to which high probability words for a given topic i have low probabilities in other topics' word distributions. The x-axis displays semantic coherence scores while the y-axis shows topic exclusivity scores for each of the eleven topic models we ran. A model with thirty topics offers the best tradeoff between the two measures, i.e. is the model closest to the upper-right quadrant.

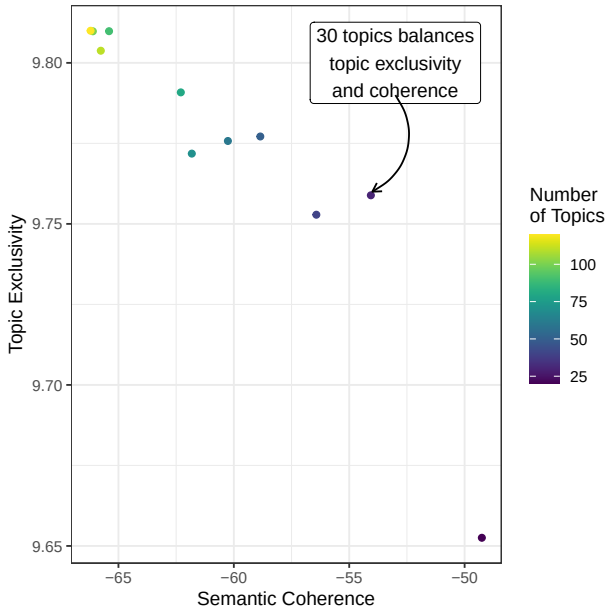

Figure A6: STM Model Topic Exclusivity vs. Semantic Coherence. This figure shows the topic exclusivity and semantic coherence scores for a range of STM models estimated on the bioweapons case study. We ultimately selected a topic model with 30 topics.

After selecting a model with thirty topics we grouped documents into clusters as defined by our topic model. Following (Roberts et al., 2020), we coarsen the quantitative document topic vectors into binary vectors, where 1 indicates a given document's topic proportion is above threshold r and 0 indicates it is below that threshold. We choose a threshold of .2 because above that threshold the number of documents unassigned to any topic bin (i.e. with no document topic proportions above .2) increases exponentially. Setting the threshold below decreases the likelihood that documents in the same bin have similar thematic features. A given document's cluster is the unordered combination of all topic bins. For example, if a document had two topics, topic 1 and 6, with greater than .2 document topic prevalence, then this document was assigned to the "1_6" topic cluster. We exclude in the binning process three "garbage" topics that we identified as substantively meaningless based on in-depth reading. These topics focused on words related to formatting in the documents.

The following plot show the count of document unassigned to any topic cluster by binning threshold. Above a threshold of .2 the number of document unassigned to any topic follows an exponential pattern before leveling off once most document are unassigned.

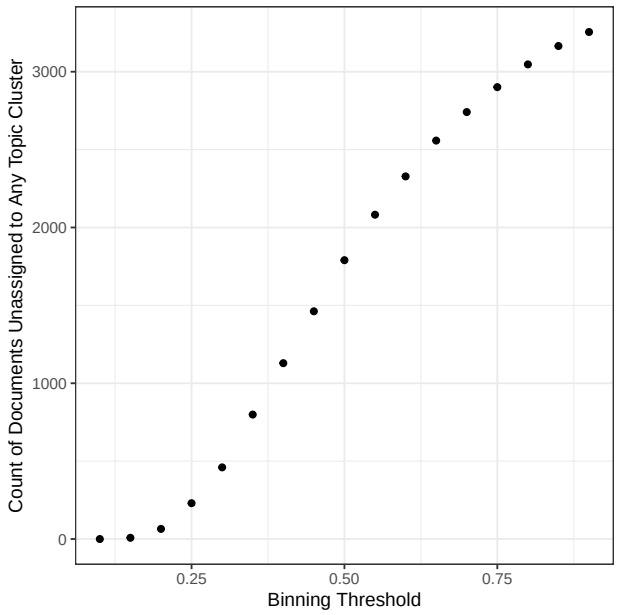

Figure A7: Count of Documents Unassigned to Any Topic Cluster by Binning Threshold. We ultimately chose a threshold of .2.

The following table includes for each topic the topic label we assigned to the topic, a falsifiable definition of each document based on reading documents with a high topic proportion for that topic,[A1] and a list of top words for each topic. We include as top words all the seven words with

[A1] We follow (Wen, Byeon, Fineman, Peskoff, and Stewart, Wen et al.) in creating a falsifiable definition for each topic.

the highest estimated probability under that topic's distribution over the corpus vocabulary as well as the seven words with the highest FREX score, the harmonic mean between a given word's frequency and exclusivity to a given topic. We exclude from this definition list the three garbage topics mentioned above.

- **Pathogens and Chemical Attacks**
  - Definition: This topic relates to pathogens and especially for the potential that Russian will use biological and/or chemical agents in its attacks on Ukraine.
  - Top Words: "can", "attack", "product", "agent", "poison", "bird", "toxic", "flu", "sarin", "bird", "factori", "substanc", "poison"
- **US admits to biological labs, Victoria Nuland**
  - Definition: This topic relates to State department official Victoria Nuland's congressional testimony that Ukraine has biological research facilities and the state department was concerned they would fall into Russian hands.
  - Top Words: "lab" , "biolab", "research", "â", "bioweapon", "pathogen" "facil", "gabbard", "lab", "biolab" "bioweapon", "nuland", "tulsi"
- **false flag, Russia conducting false flag, Ukranine conducting false flag, false pretexts for war**
  - Definition: This topic relates to claims by both Ukraine and Russia that the other was going to stage a "false flag" attack to create a false pretext for retaliation.
  - Top Words: "accus", "fals", "claim", "alleg" "oper", "attack", "un", "psaki", "flag", "fals", "accus", "un", "pretext", "jen"
- **Gardening, Agriculture, Sleep, Daily Needs**
  - Definition: This topic relates to daily needs in Ukraine during the war, especially agriculture and sleep.
  - Top Words: "can", "water", "time", "need", "bodi", "scientist", "sea" "water", "sea", "tree", "sleep", "wast", "brain", "scientist"
- **Drones, aircrafts, military equipment**
  - Definition: This topic deals with military equipment and conventional weapons.
  - Top Words: "equip", "system", "defens", "forc", "provid", "air", "drone", "armor", "drone", "equip", "vehicl", "aircraft", "system", "helicopt"

- **Nuclear attack**
  - Definition: This topic deals with nuclear weapons and the threats of a nuclear attack.
  - Top Words: "nuclear", "attack", "forc", "power", "missil", "escal", "destruct", "nuclear", "chernobyl", "escal", "arsenal", "tactic", "strike", "scenario"
- **Civilians, attacks on civilians**
  - Definition: This topic deals with the impacts of conflict on civilian populations.
  - Top Words: "citi", "civilian", "kyiv", "forc", "peopl", "mariupol", "attack", "citi", "mariupol", "zelenskyy" "civilian", "town", "kyiv", "bomb"
- **FSB, Russian security services, poisoning attack**
  - Definition: This topic primarily deals with activities by Russian security services, including a poisoning attack by negotiators during the conflict.
  - Top Words: "oper" , "moscow", "special", "negoti", "donbass", "republ", "kiev", "donbass", "lugansk", "kiev", "republ", "negoti", "demilitar", "donetsk"
- **Satellite data, sharing meterological data**
  - Definition: This topic deals with a controversy over western organizations sharing satelitte data with Russia.
  - Top Words: "european", "eu", "data", "union", "organ", "europ", "minist", "eu", "european", "union", "data", "rubl", "meteorolog", "satellit"
- **Online misinformation, fact checking**
  - Definition: This topic deals with allegations of misinformation and conspiracy theories in media content.
  - Top Words: "media", "news", "claim" "post", "conspiraci", "social", "theori", "conspiraci", "theori", "user", "platform", "outlet", "post", "narrat"
- **Biden official statements, Biden visit to Ukraine**
  - Definition: This topic relates to official statements and actions by Biden and his administration.
  - Top Words: "biden", "presid", "mr", "offici", "putin", "nation", "hous", "mr", "sullivan", "wednesday", "thursday" "biden", "poland", "jake"

- **Russian investigation into biolabs, statements on biolabs**
  - Definition: This topic relates to official statements from Russian authorities on the biolabs as well as the Russian commission investigating the biolabs.
  - Top Words: "laboratori", "ministri", "activ", "defens", "note", "foreign", "repres", "tass", "military-biolog", "laboratori", "convent", "duma", "un", "zakharova"
- **Allegation of Russian propaganda**
  - Definition: This topic relates to allegations that Russia is spreading propaganda.
  - Top Words: "feder", "fake", "alleg" "propaganda", "inform", "spread", "laboratori", "fake", "propaganda", "biolaboratori", "propagandist", "feder", "alleg", "spread"
- **NATO summit, NATO statements**
  - Definition: This topic focuses on NATO summits and statements from NATO officials.
  - Top Words: "nato", "allianc", "support", "summit", "stoltenberg", "alli", "general", "allianc", "nato", "summit", "stoltenberg", "brussel", "alli", "assist"
- **COVID-19, pandemic, vaccines**
  - Definition: This topic deals with COVID-19 related news and information.
  - Top Words: "vaccin", "health", "coronavirus", "new", "peopl", "covid-19", "pandem", "vaccin", "coronavirus", "covid-19","omicron", "covid", "pandem", "dr"
- **Russian expansionism, global order, collapse of global systems**
  - Definition: This topic focuses on discussions of large scale global order and what threatens it. Examples include discussions of Russian expansionism/revisionism and the threat of growing global monopolies.
  - Top Words: "world", "peopl", "even", "now", "can", "power", "polit", "globalist", "societi", "soviet", "collaps", "freedom", "simpli", "polit"
- **US domestic politics**
  - Definition: This topic focuses on US national domestic politics.
  - Top Words: "senat", "republican", "trump", "presid", "democrat", "biden", "court", "republican", "jackson", "judg", "suprem", "democrat", "trump", "senat"
- **Hunter Biden**

- Definition: This topic relates to Hunter Biden's involvement in Ukraine.
- Top Words: "biden", "hunter", "presid", "fund", "metabiota", "compani", "son", "hunter", "metabiota", "laptop", "son", "invest", "rosemont", "seneca"

- **Biolabs research**
  - Definition: This topic refers to research being done in the biological research facilities in Ukraine.
  - Top Words: "laboratori", "ministri", "defens", "research", "document", "kirillov", "pathogen", "kirillov", "project", "sampl", "document", "igor", "studi", "bat"

- **families, children**
  - Definition: This topic relates to stories about families and children.
  - Top Words: "children", "famili", "parent", "child", "mother", "now", "hous", "parent", "mother", "daughter" "child", "children", "girl", "babi"

- **Holidays**
  - Definition: This topic focuses on holidays and festivals.
  - Top Words: "day", "year", "februari", "first", "world", "peopl", "intern", "celebr", "holiday", "writer", "artist", "spring", "film", "cancel"

- **Fox news, Tucker Carlson, right wing media**
  - Definition: This topic is mostly focused on Fox news shows such as Tucker Carlson.
  - Top Words: "carlson", "like", "peopl", "know", "just", "go", "one", "carlson", "tucker", "clip", "yes", "fox", "male", "tonight"

- **Statements from Ukranian or US leadership about Russian leadership**
  - Definition: This topic focuses on statements from US or Ukranian leadership concerning Russian leadership.
  - Top Words: "putin", "presid", "vladimir", "minist", "may", "attack" "kremlin", "putin", "vladimir", "lavrov", "zelenski", "kremlin", "minist", "rbc-ukrain"

- **China's statements on Ukraine**
  - Definition: This topic focuses on statements from Chinese leadership and officials on the conflict in Ukraine.

- Top Words: "china", "chines", "beij", "support", "xi", "offici", "call", "xi", "beij", "chines", "china", "china'", "zhao", "jinp"

- **energy, gas, inflation from invasion**
  - Definition: This topic relates to the price of gas and commodities and concerns over inflation from the conflict.
  - Top Words: "gas", "price", "oil" "energi", "$", "new", "product", "oil", "energi", "price", "gas", "inflat", "climat", "fuel"

- **Attacks on civilians**
  - Definition: This topic relates to concerns over especially Russian attacks on civilians.
  - Top Words: "region", "feder", "forc", "occupi", "arm", "march", "enemi" "region", "enemi", "occupi", "provoc", "shell", "mariupol", "villag"

- **Legal, investigation**
  - Definition: This topic relates to legal issues and investigations as well as redress.
  - Top Words: "law", "investig", "person", "polic", "crimin", "crime", "offic", "household", "crimin", "polic", "prison", "green", "law", "prosecutor"

D.2 Additional Analyses

Figure A8: Percent of Low Quality and Popular News Articles sharing narratives with Russian state media (left) and Ukranian media (right) by topic. Articles were assigned to topics if their estimated prevalence for a given topic was greater than .2 or 20%.

E Additional Validation Results

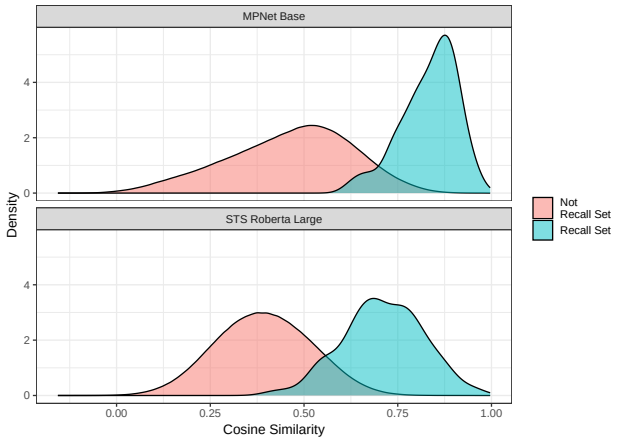

Figure A9: Distribution of Between Article Cosine Similarity Scores based on Embedding Modles (MPNet Base top, STS Roberta bottom). We see greater separation between known positive cases (in blue) for MPNet Base than STS Roberta.